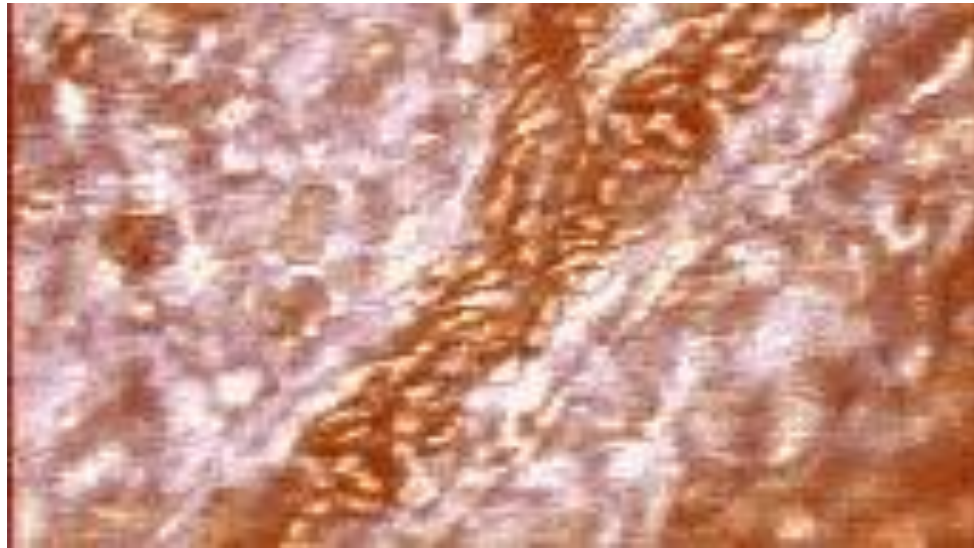


The Physiology of Blood



Introduction

Outline

- Characteristics and Functions of blood
- Blood composition
 - Formed elements of the blood
 - Plasma proteins and their functions
- Red blood cells: function and count
 - Erythropoiesis and sites of RBC formation
 - Hemoglobin: synthesis, types
 - RBC disorders
 - Anemia, types
 - Polycythemia, types

Outline

- White blood cells (Leukocytes)
 - Types and functions of WBCs
 - Hemopoiesis
 - Cytokines: regulation of WBC production
 - Immunity
- Platelets
 - Hemostasis (Blood Coagulation)
- Blood Groups & Blood Transfusion

Introduction

- Blood is the body's only fluid connective tissue
- It's a sticky, opaque fluid with a metallic taste
- pH of blood is 7.35–7.45 and its salt content is 0.9%
- Temperature is 38°C, slightly higher than “normal” body temperature

Introduction

- Characteristics of blood:
 - **Color**
 - Bright red – oxygenated blood – arterial blood
 - Dark red – deoxygenated blood – venous blood
 - **Viscosity**
 - Blood is 3 times thicker and denser than pure water
 - **Volume:**
 - Male – 5-6 litre
 - Female – 4-5 litre
 - **Density (specific gravity) → Whole blood = 1.060**

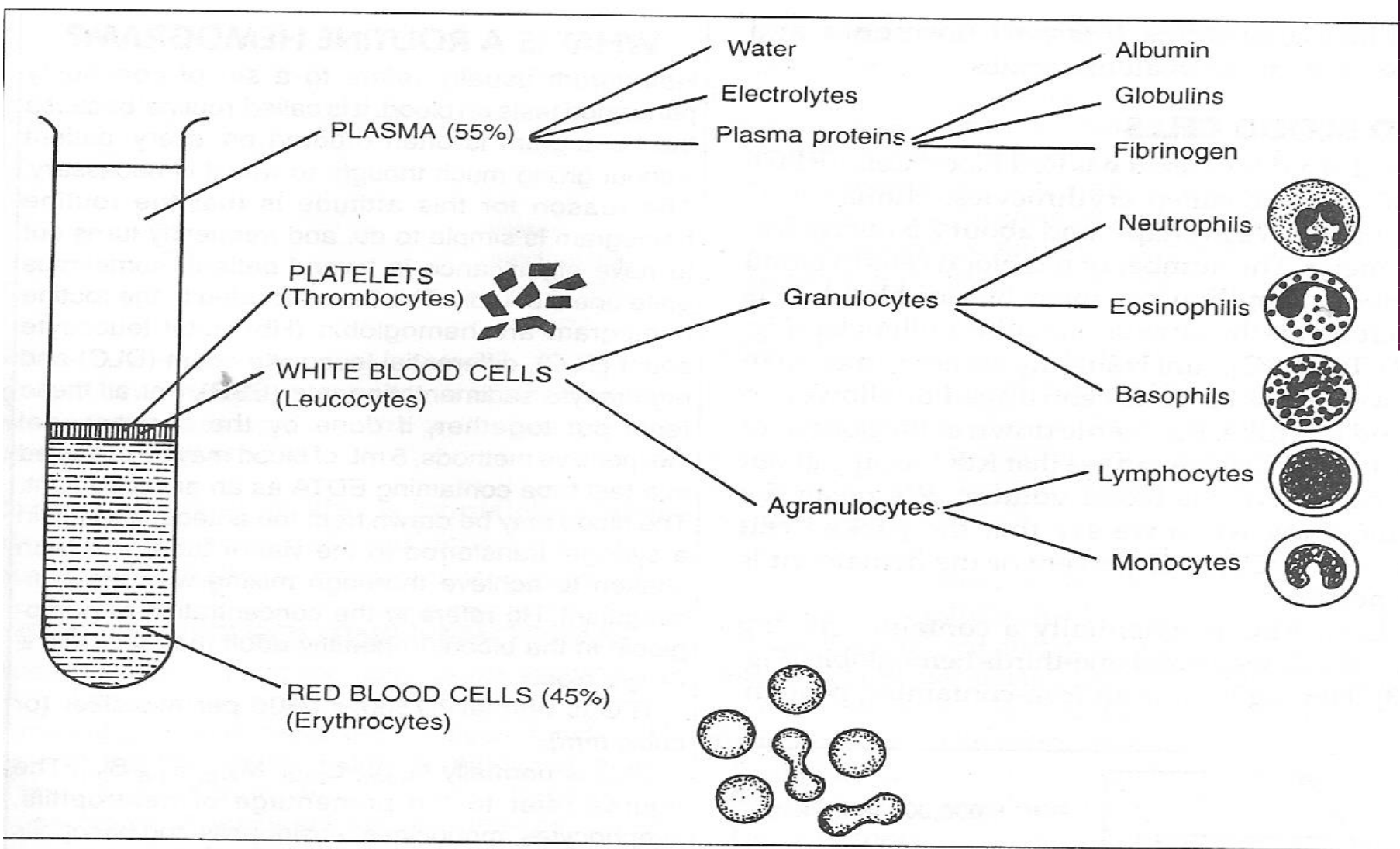
Functions of Blood

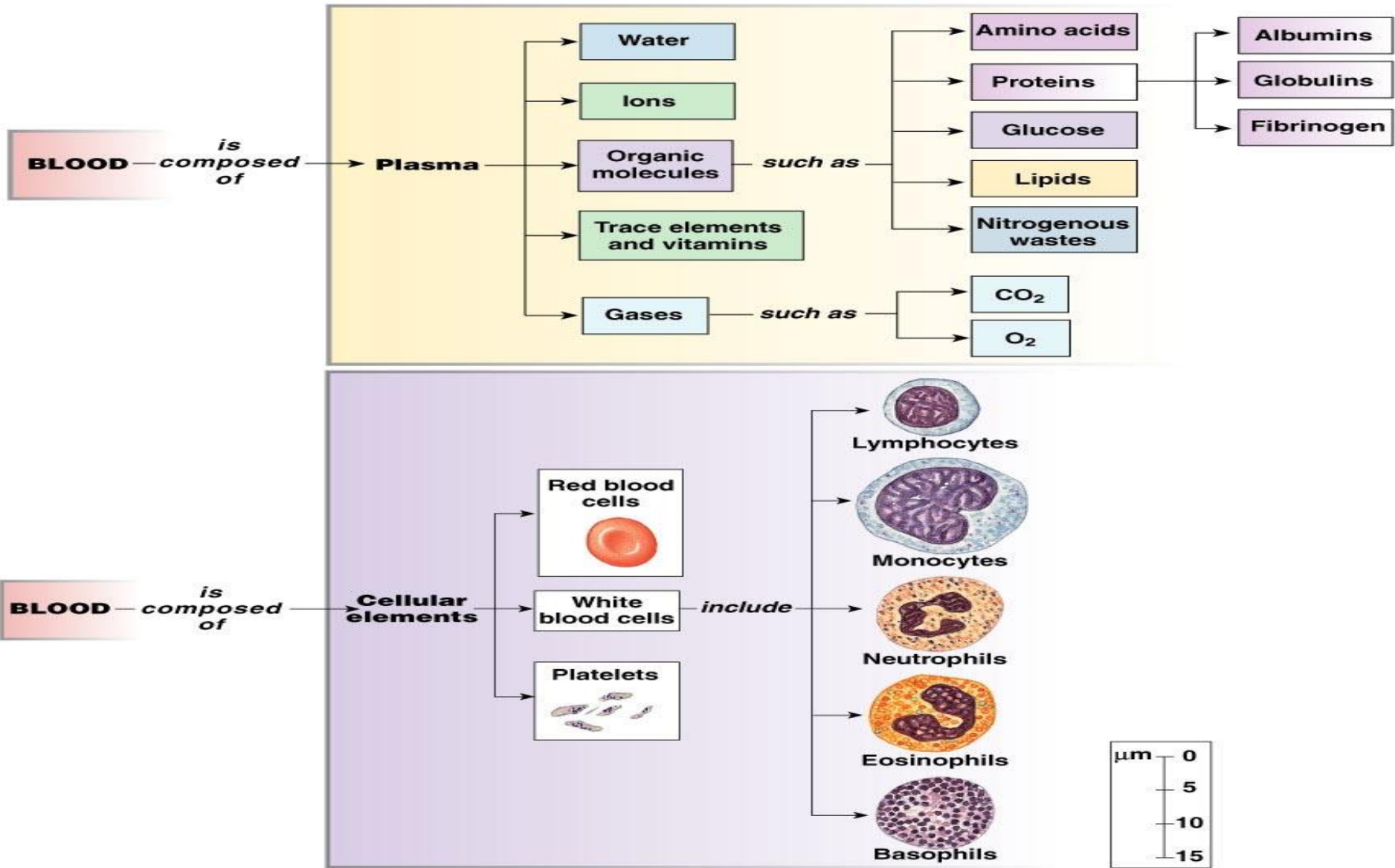
- **Transport:**
 - Oxygen from the lungs and nutrients from the digestive tract
 - Metabolic wastes from cells to the lungs and kidneys for elimination
 - Hormones from endocrine glands to target organs
- **Regulation:**
 - Body temperature by absorbing and distributing heat
 - Acid-base balance (pH regulation) using buffer systems (e.g., Hemoglobin, HCO_3^-)
 - Adequate fluid volume in the circulatory system

Functions of Blood

- **Hemostasis:**
 - Prevents blood loss by:
 - Activating plasma proteins and platelets
 - Initiating clot formation when a vessel is broken
- **Immunity:**
 - Prevents infection by:
 - Synthesizing and utilizing antibodies
 - Activating WBCs to defend the body against foreign invaders

Composition of Blood

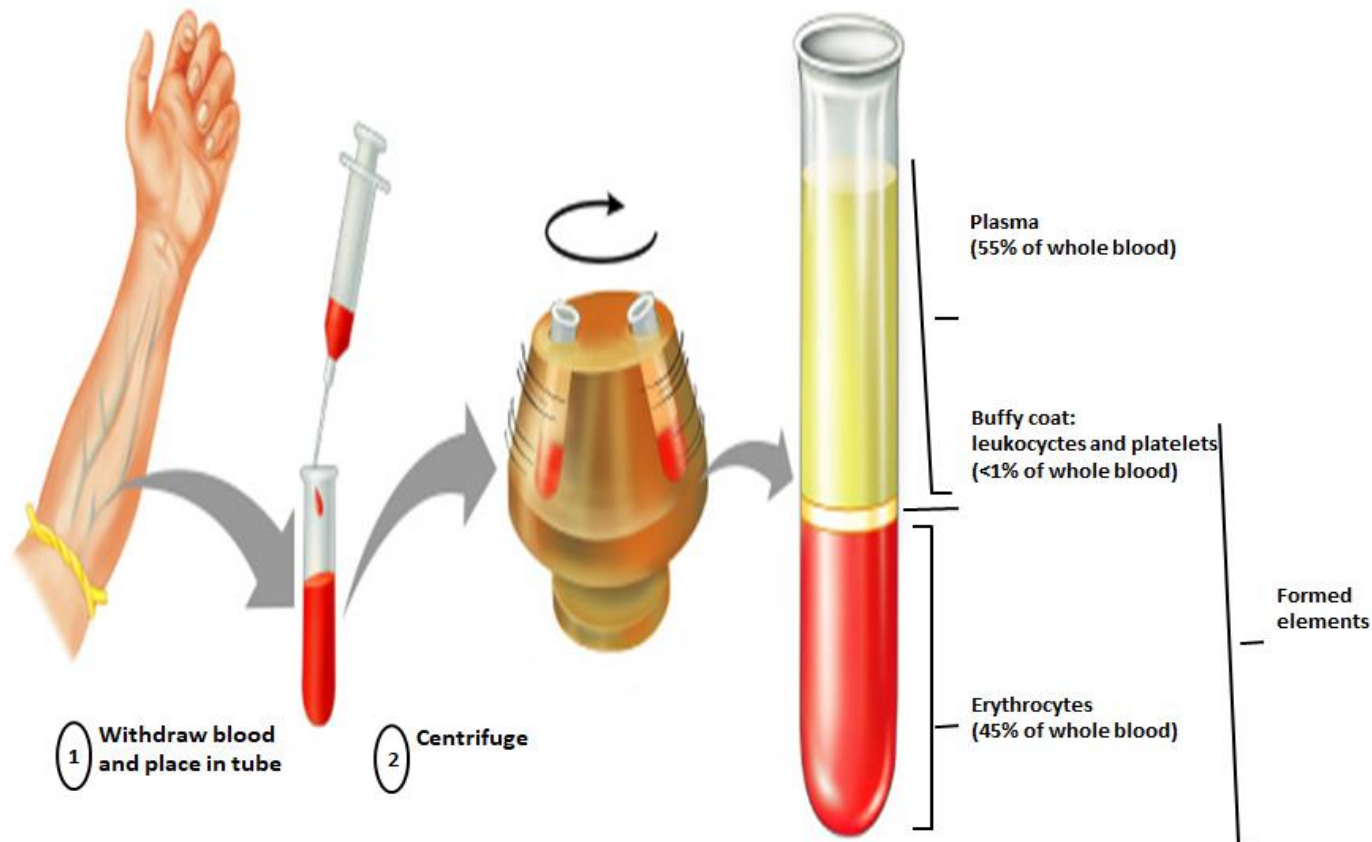




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Hematocrit (Hct)

- Proportion of packed cell volume (RBCs)
- Hct increases in dehydration & polycythemia and decreases in anemia



Serum vs. Plasma

- **Serum** is the yellowish fluid that forms after blood is left to clot
- Serum has similar composition to plasma except that its fibrinogen have been removed
- **Plasma** is the liquid that remains when clotting is prevented with the addition of an anticoagulants (Heparin, EDTA...)
- **Serum** is the liquid that remains after the blood has clotted.

Plasma

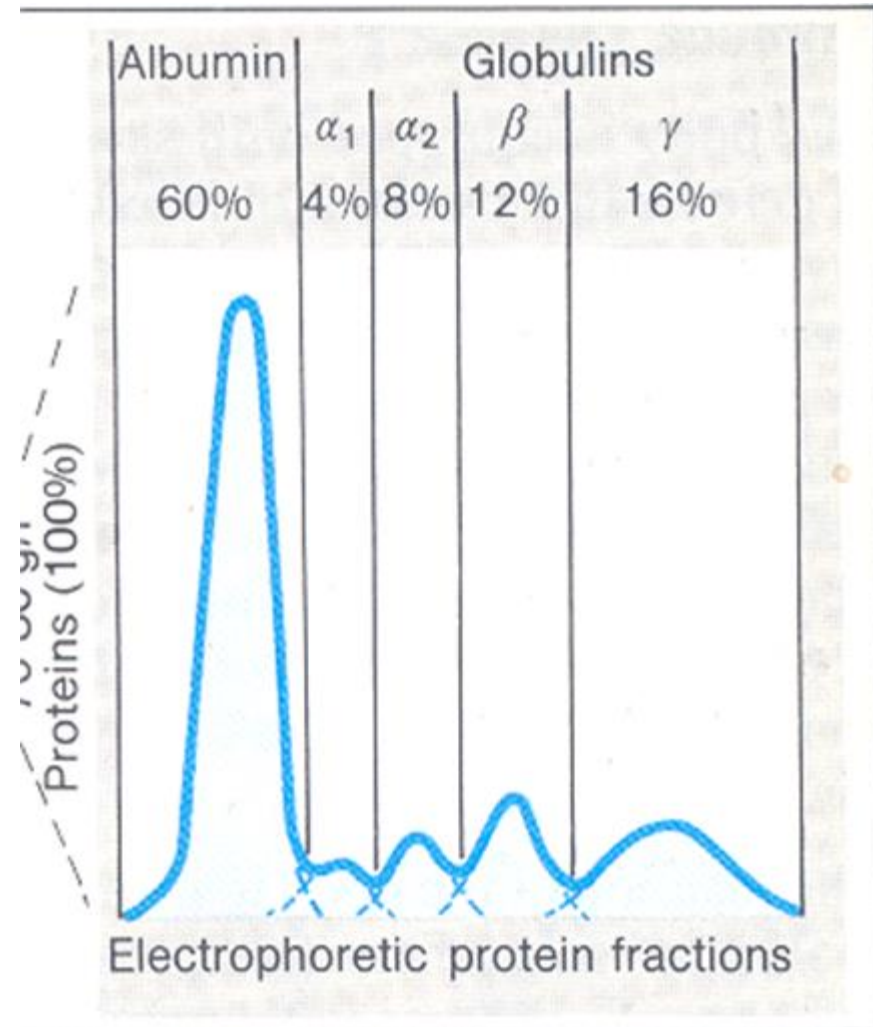
- Liquid portion of the blood
- Constitutes 55% of the blood volume
- Osmolality - 300 mosm/l
- Composition of plasma:
 - Water (90%)
 - Inorganic constituents: electrolytes (1%)
 - Respiratory gases – oxygen and carbon dioxide
 - Organic constituents (9%)

Plasma

- Organic constituents
 - Plasma proteins: albumin, globulins, clotting proteins and others (7%)
 - Lipids, lipoproteins, phospholipids
 - Hormones and enzymes
 - Nutrients: CHO, vitamins, amino acids, fats
 - Metabolic waste products: urea, creatinine

Plasma Proteins

- TPP $\rightarrow$ 7 g/dl
- 3 principal plasma proteins:
 - Albumin $\rightarrow$ 4 g/dl
 - Globulin $\rightarrow$ 2.7 g/dl
 - Fibrinogen $\rightarrow$ 0.3 g/dl
- Plasma proteins are synthesized by hepatocytes, lymphocytes, platelets and endothelial cells



Plasma Proteins

- **Albumin**
 - Produced in the liver
 - Responsible for Colloid osmotic pressure (**oncotic pressure**)
 - This force helps to move plasma back into the blood and thus keeps blood volume to normal level
 - It also transports nutrients (lipids, hormones, bilirubin...)

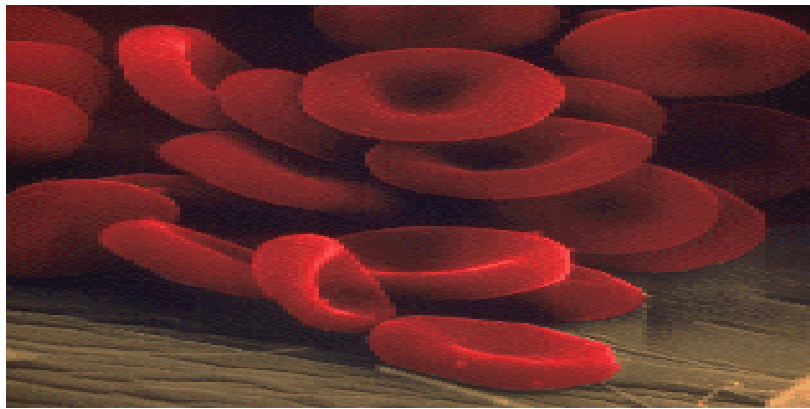
Plasma Proteins

- Globulin
 - 3-types → alpha1&2, beta, and gamma globulins
 - α & γ are made in the liver and transport lipoproteins, iron, hormones...
 - γ -globulins act as antibodies
- Fibrinogen
 - Synthesized in the liver and involved in blood clotting

Plasma Proteins

- **Functions:**
 1. Immunologic function: γ -globulins are immunoglobulin's (antibodies)
 2. Haemostasis: fibrinogen, prothrombin and most other clotting factors are plasma proteins
 3. Energy source during starvation
 4. Transport of hormones, electrolytes and drugs
 5. pH regulation (buffering function)
 6. Maintenance of plasma osmotic pressure (mainly by albumin)
 7. Enzymatic and hormonal function

The Physiology of Blood

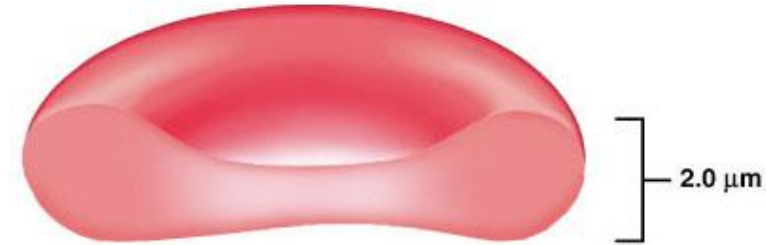


Red Blood Cells (Erythrocytes)

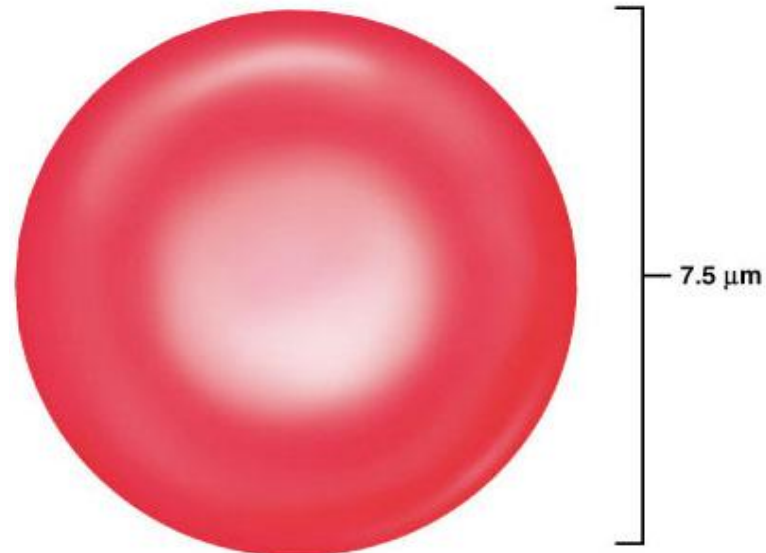
Erythrocytes

- **Characteristics:**

- No nucleus, no organelles
- Biconcave disk, changed as necessary
- 7.5 μm diameter and 2 μm thickness
- ATP is generated anaerobically
- Major content of RBCs is Hb (97%)
- RBC Count:
 - Male = 5-6 (5.2) millions/ mm^3
 - Female = 4-5 (4.6) millions/ mm^3



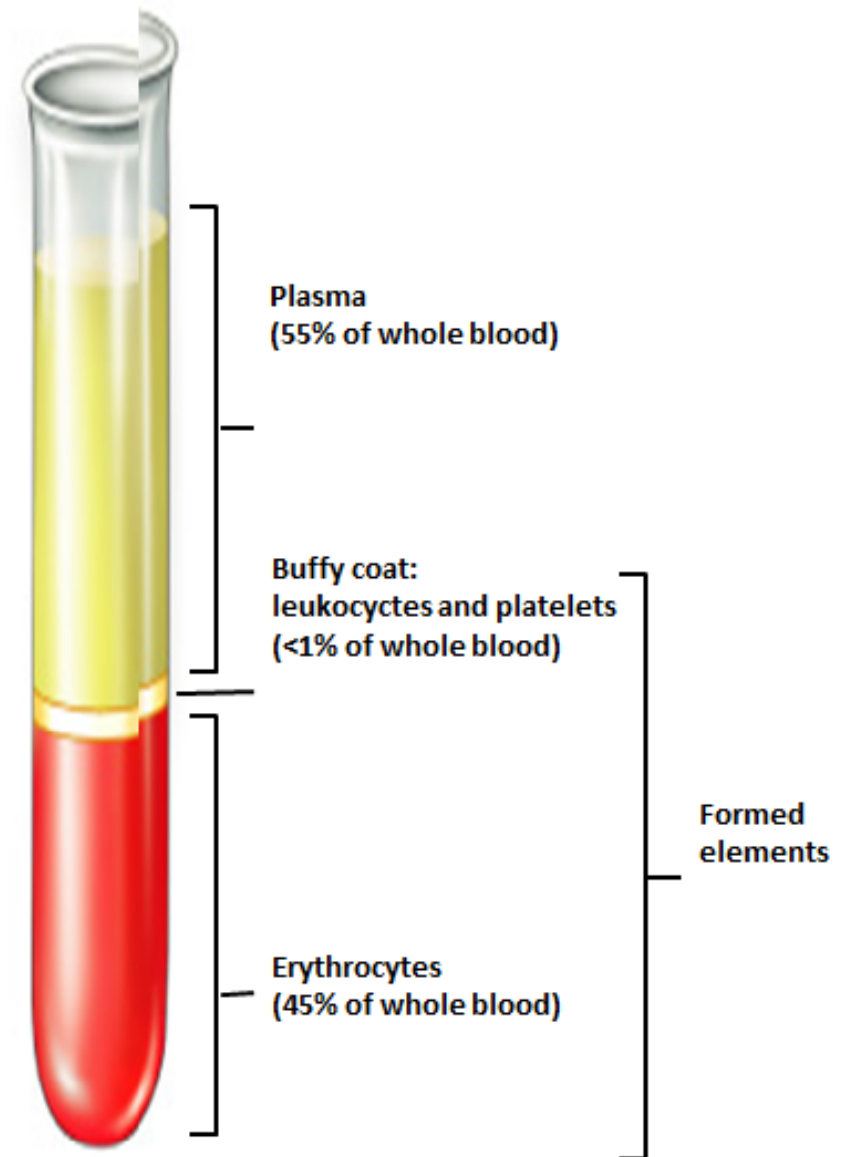
Side view



Top view

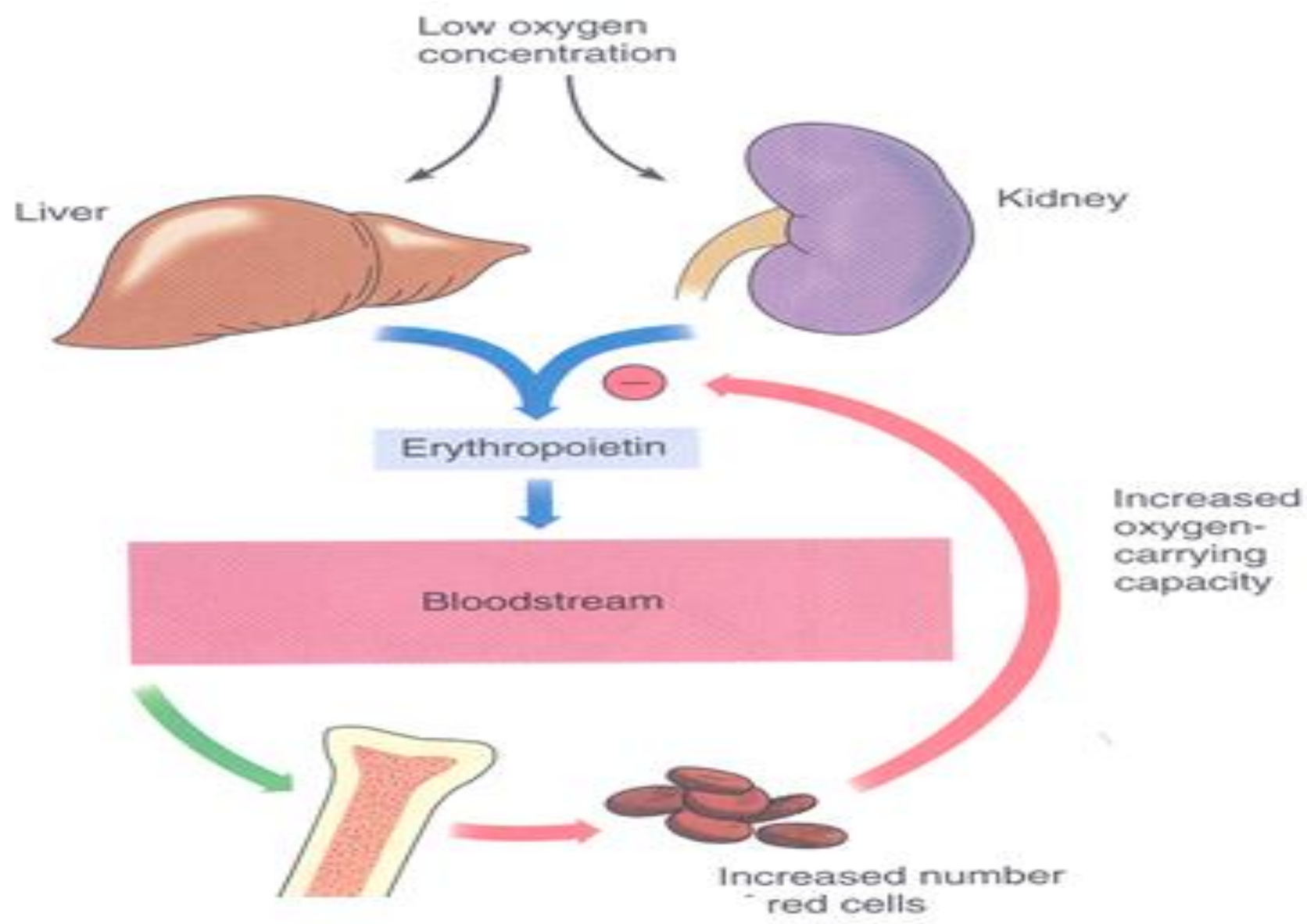
Erythrocytes

- **Characteristics:**
 - Hematocrit:
 - M = 42-48% (45%)
 - F = 38-43% (42%)
 - Hemoglobin (Hb)
- **Function:**
 - Transport of O_2 and CO_2
 - Regulation of acid-base balance

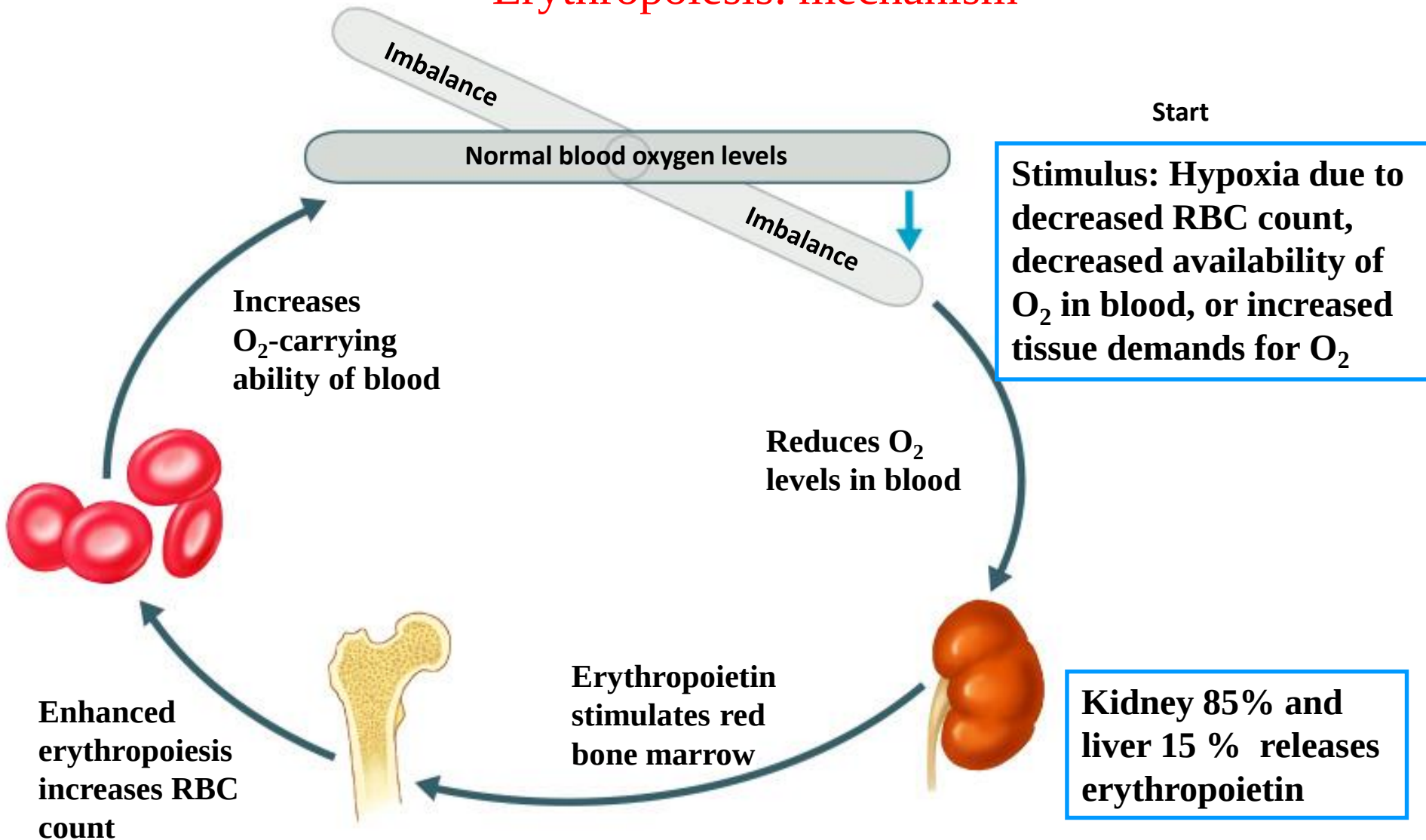


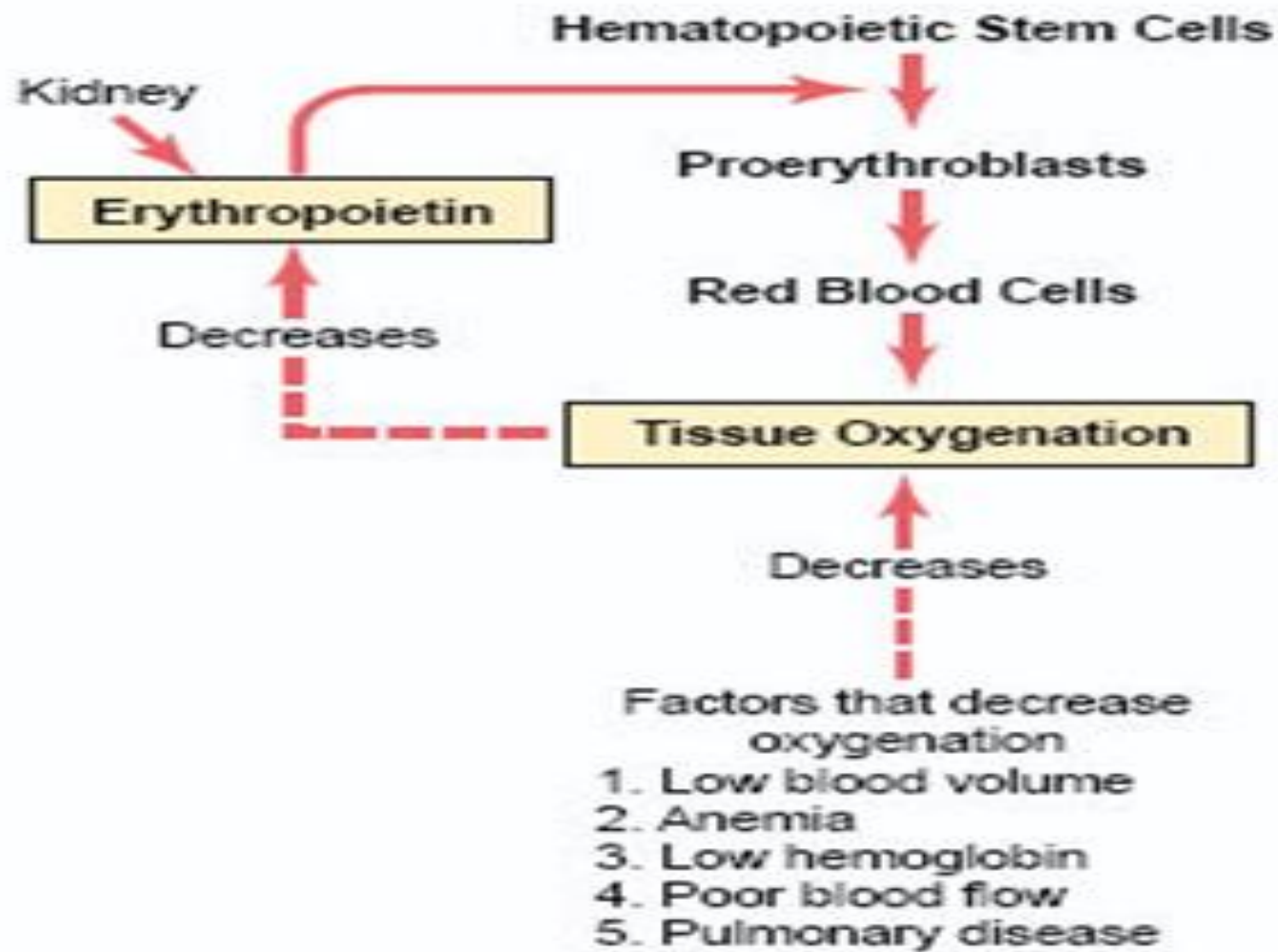
Erythropoiesis

- Erythropoiesis – occurs by negative feedback mechanism:
 - Hypoxia stimulates kidney → kidney cells release erythropoietin hormone → Erythropoietin reaches bone marrow through blood and induce stem cells to produce more RBC's → RBC's release O₂ to the body
- The process takes about 5-days
- Liver plays a critical role in RBC formation: site of hemoglobin synthesis, storage area of iron and vit-B12, also secretes erythropoietin
- Kidney and Liver insufficiency can cause anemia, b/s of lower synthesis of erythropoietin



Erythropoiesis: mechanism





Erythropoiesis

- Regulation

- Normally, the number of circulating RBCs remains constant → reflects a balance between RBC production and destruction
- Too few red blood cells leads to tissue hypoxia
- Too many red blood cells causes increased blood viscosity
- Erythropoietin (EPO) from JG-cells and hepatocytes
- Iron, vit-B12, folic acid...
- Intrinsic factor from parietal cells of the stomach

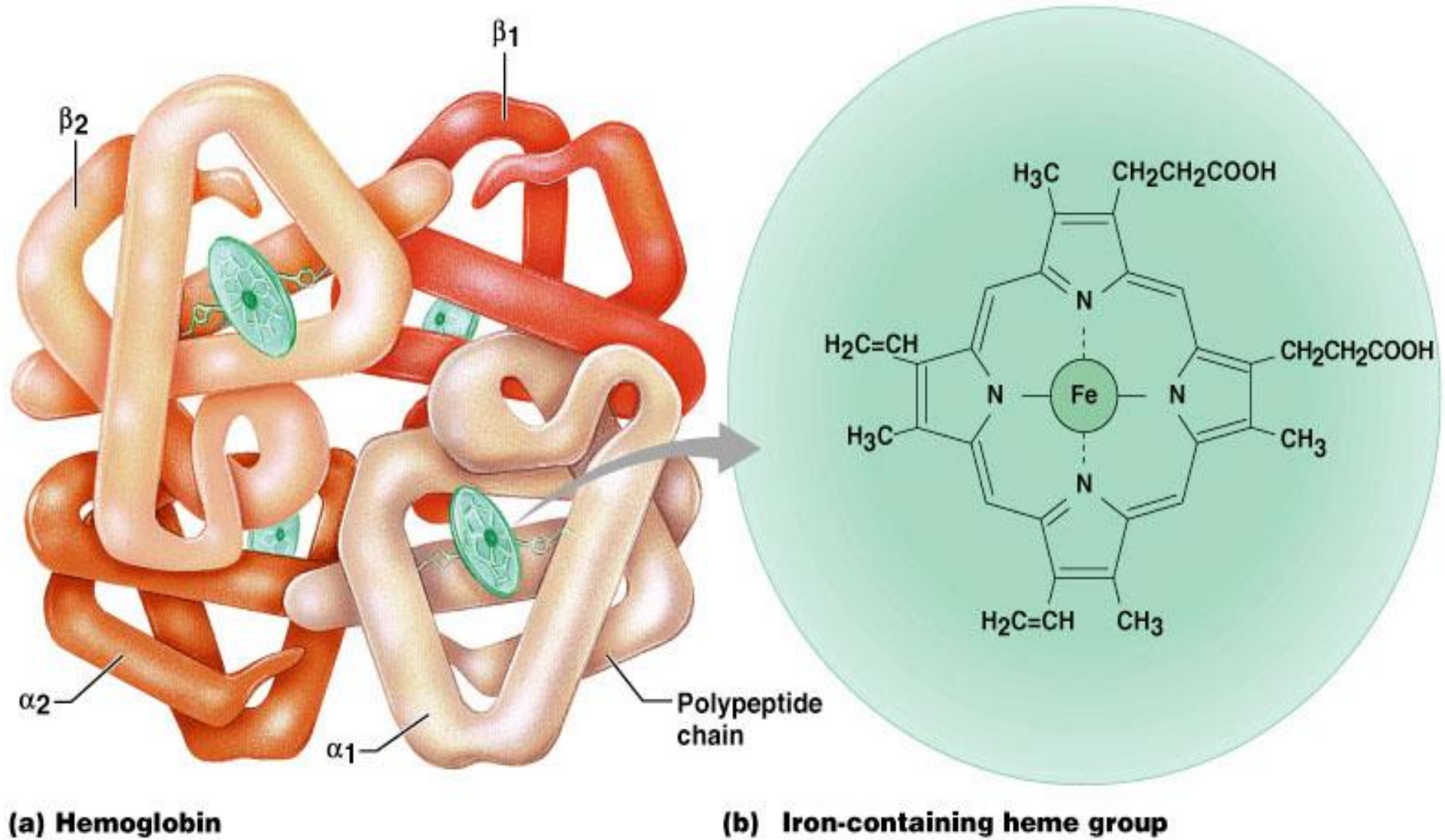
Erythropoiesis

- Vitamin B-12 and Folic acid
 - Important for DNA synthesis and thus cell division of RBC
 - Deficiency causes pernicious anemia/ megaloblastic anemia
 - RBCs become megaloblastic (*impaired DNA synthesis, which inhibits nuclear division*)
 - Pernicious anemia is mainly caused by the absence of production of **Intrinsic factor** in the stomach → which increases Vit-B-12 absorption

Hemoglobin

- Function: transport of O₂ and CO₂, and as a buffer to maintain acid-base balance
- Normal concentration: 16 g/dl in Male and 14 g/dl in Female
- Structural composition
 - Globin → protein with 4 polypeptide chains (2 α & 2 β)
 - Heme → Fe²⁺ containing central part to which oxygen binds

Structure of Hemoglobin

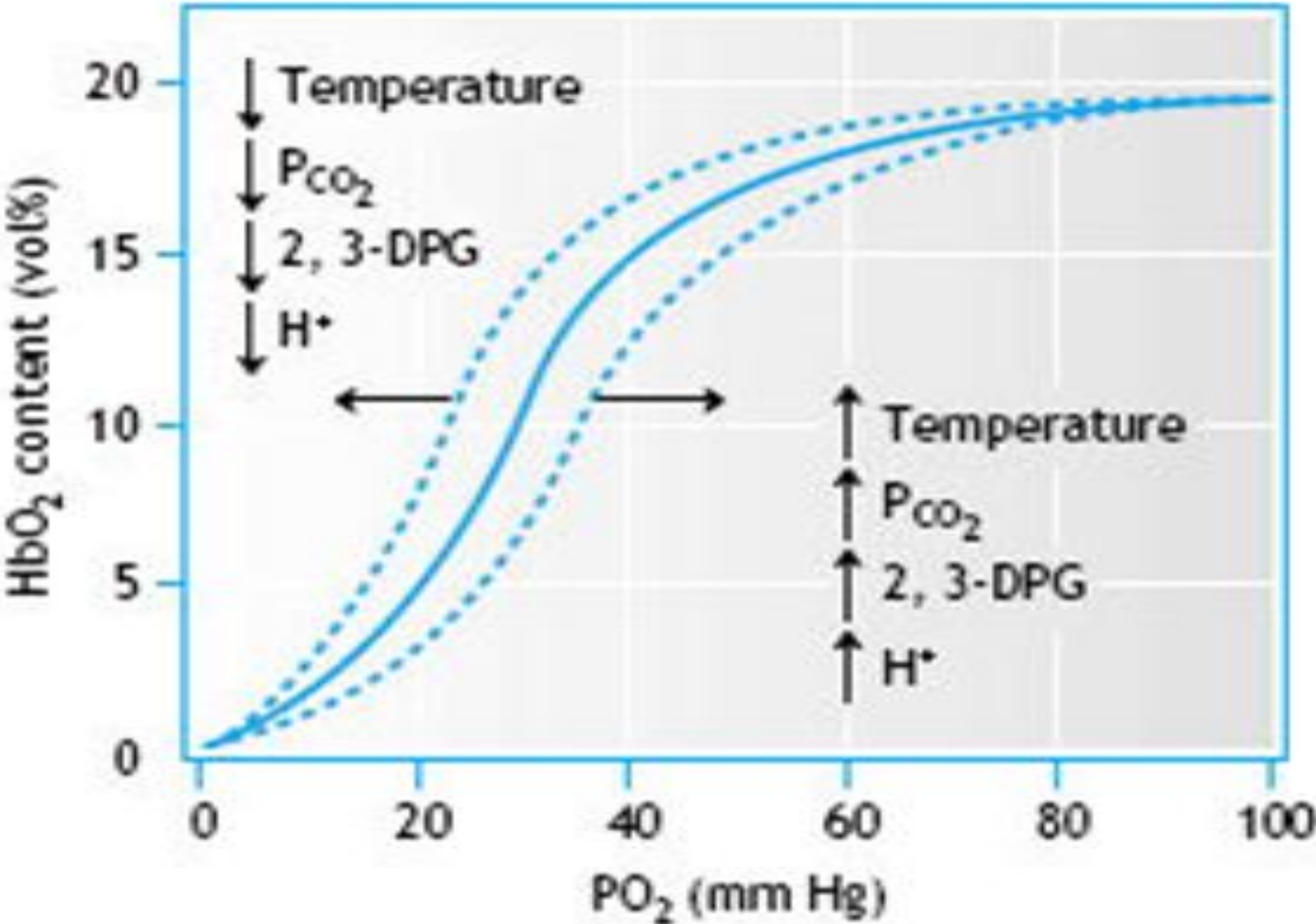


Hemoglobin

- Each polypeptide has one Heme group
- Each Heme contains one Fe^{2+} that carries 1- O_2 molecule
- So, totally, 4- O_2 molecules are carried by 1-Hb molecule
- 1g Hb binds with 1.34 ml O_2
- So, 15g Hb x 1.34 ml O_2 = 20.1 ml of O_2 /100 ml blood

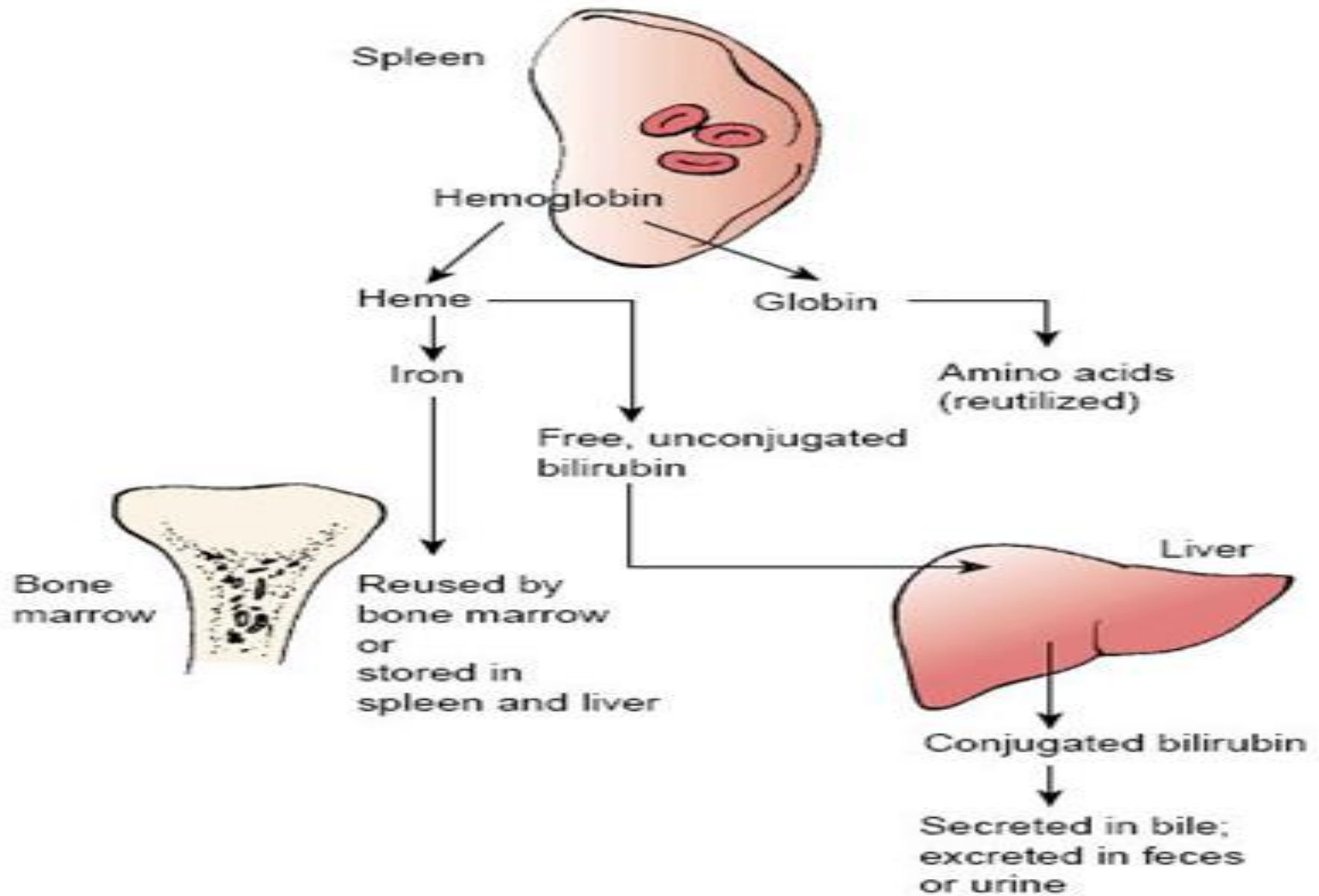
Hg-O₂ Saturation Curve

- Hb combines loosely and reversibly with O₂
- Hence, it gives-off O₂ easily to tissues in hypoxic conditions and Loads O₂ easily from the lung tissues when available
- Factors favoring Hb-O₂ loading (shift the curve to the left)
 - Decrease in Temperature, PCO₂, H⁺ and 2,3- DPG
- Factors favoring Hb-O₂ unloading (shift the curve to the right)
 - Increase in Temperature, PCO₂, H⁺ and 2,3- DPG



Destruction of Erythrocytes

- The life span of an erythrocyte is 120 days
- Old erythrocytes become rigid and fragile, and their hemoglobin begins to degenerate
- Dead erythrocytes are engulfed by macrophages
- Heme and globin are separated and the iron is salvaged for reuse
- The **heme** part is converted into **bilirubin**
- **Bilirubin** is the main component of biliary secretion



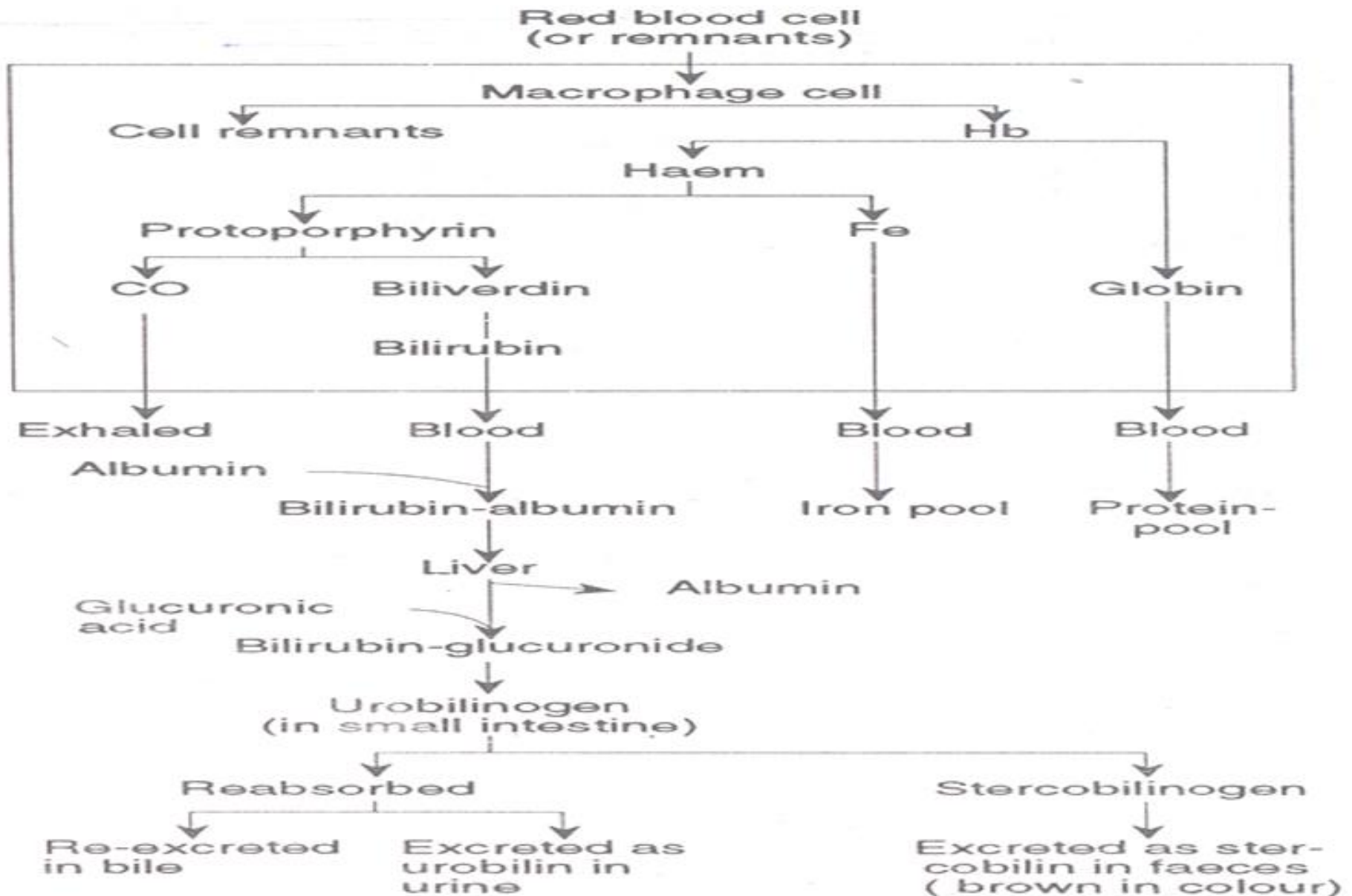
Jaundice

- Yellow coloration of the skin and sclera
- Due to excessive bilirubin in plasma
- Normal plasma bilirubin level is 0.5 mg/dl
- In case of jaundice, bilirubin level is elevated up to 40 mg/dl
- The skin and the sclera look yellow when bilirubin level rises >1.5 mg/dl



Jaundice

- Causes/types of jaundice
 1. Hemolytic jaundice: ↑RBC destruction
 2. Obstructive jaundice:
 - Obstruction of bile ducts
 - Damage of hepatocytes

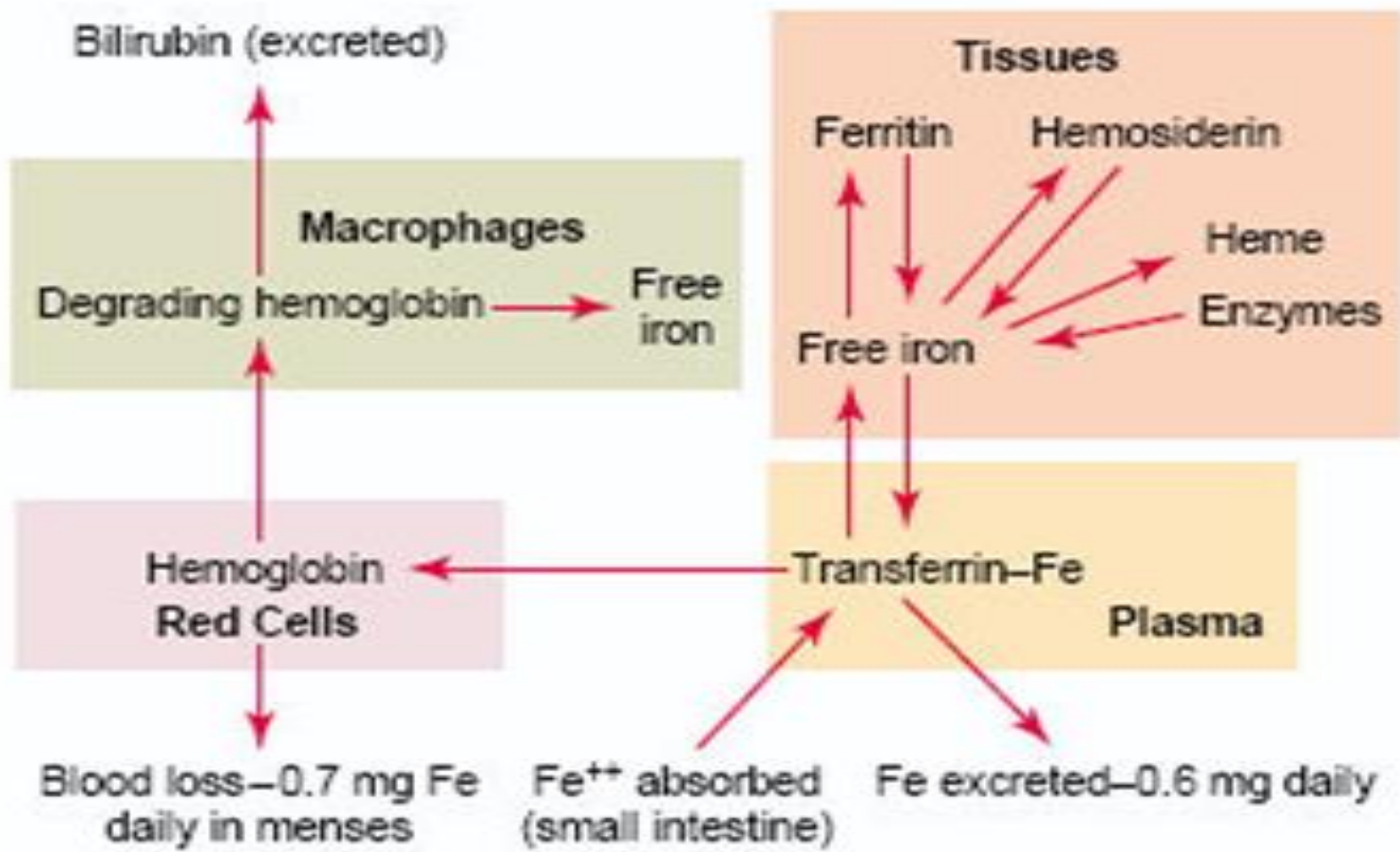


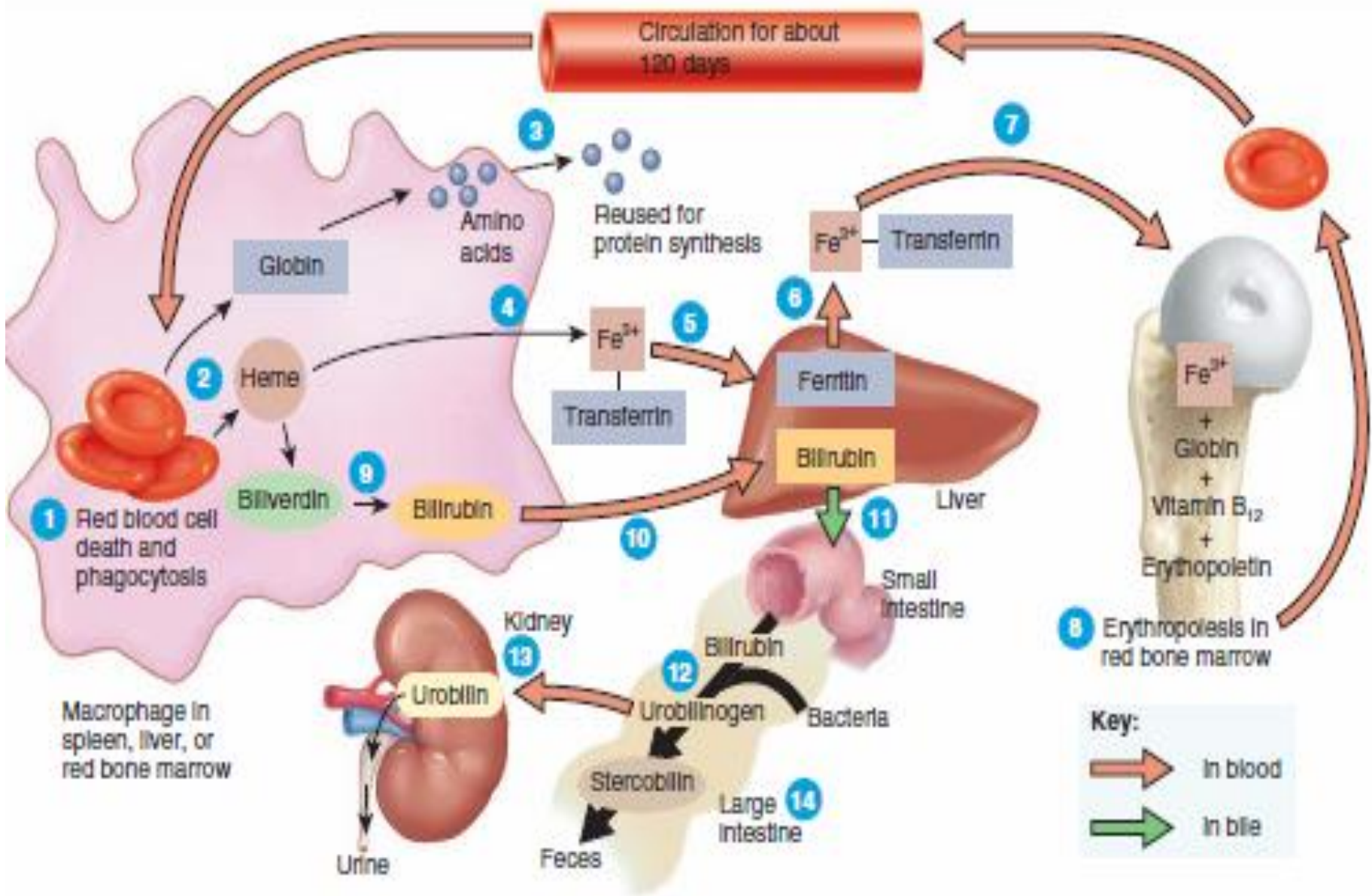
Iron Metabolism

- Iron is essential for the formation of Hemoglobin and Myoglobin
- Serve as a **cofactor** for cytochrome enzymes, peroxidases and catalases
- A normal adult contains a total of **4 g of iron** and **needs 1 mg/day**
- **Transport**
 - Iron is transported through the blood plasma bound to **Transferrin**
- **Absorption**
 - Ferrous form (Fe^{2+}) is better absorbed than ferric form (Fe^{3+})
 - **Ascorbic acid** facilitates absorption of ferrous

Iron Metabolism

- **Storage:**
 - Transferrin to liver and in the liver it combines with **apo-ferritin** for storage, and can be given to tissues in need
 - Iron is also stored as **hemosiderin** (insoluble form)
- **Iron exists in the body in different forms**
 - 65% in the form of Hb
 - 4% in the form of Mb
 - 1% in the form of cytochromes
 - 0.1% in combination with transferrin
 - 15-30% stored in the liver in the form of apo-ferretin and hemosiderin





Erythrocyte Disorders

- Includes:
 - Anemia
 - Polycythemia
- **Anemia**
 - A decrease in the hemoglobin level, RBC count or both
 - Anemia leads to a decrease in the O₂ carrying capacity of the blood
 - Generally anemia results from
 - Excessive blood loss (bleeding)
 - RBC destruction (hemolysis)
 - Decreased production of RBC (lack of nutrition) in the circulation

Anemia

1. Blood loss (hemorrhagic) anemia

- Result of acute or chronic blood loss
- The Fe^{2+} absorbed from the intestine is not enough to produce adequate Hemoglobin to replace the lost Hemoglobin that took place during bleeding
- As the result, the RBC's produced are smaller in size and have lower Hb level causing a **microcytic, hypochromic anemia**

Anemia

2. Aplastic anemia

- Resulted from **destruction of bone marrow**
- Can be caused by irradiation, drugs, chemicals and leukemia
- Effect is **lower RBC production**

3. Iron deficiency anemia

- Common in infants and females
- Results from: Inadequate intake of iron-containing foods and
Impaired iron absorption

Anemia

4. Pernicious anemia

- caused by:
 - Inability to absorb **Vit-B12** and **folic acid** from the intestine
 - Absence of intrinsic factor (gastrectomy, gastric atrophy)
 - Impaired DNA synthesis → RBCs can not proliferate properly
 - RBC's become larger and oversized (megaloblastic) and have fragile membranes that easily rupture and decrease their number

Anemia

5. Hemolytic anemia

- Prematurely ruptured erythrocytes or formation of fragile RBCs due to sickle cell, thalassemia (a hereditary form of anemia), transfusion reaction, snake venoms, malaria, **erythroblastosis fetalis**

a. Hereditary spherocytosis:

- Spherical shaped RBC (not biconcave) that rupture easily

b. Erythroblastosis fetalis:

- Destruction and decrease of RBC number of the fetus by antibodies from Rh-negative mother

Anemia

4. Hemolytic anemia

c. Sickle cell anemia:

- Caused by substitution of valine by glutamic acid in the beta chain of Hb
- This defect causes RBCs to become sickle-shaped after oxygen is released to tissue
- RBC's become rigid and can get stuck in blood vessels
- When exposed to low O₂ (hypoxia), Hb precipitates forming long crystals and damaging the cell-membrane

Sickle RBC



Normal RBC



Polycythemia

- Production of too much RBC (RBC count > 6 million/mm³)
- Can be primary or secondary polycythemia

1. Primary polycythemia (Polycythemia Vera (erythremia))

- Pathological cause
- Mainly due to genetic defect as in Tumorous or cancerous production of RBC – over activity of bone marrow
- About 7-8 million RBC/ mm³
- Moreover, increases are also seen in WBC and platelet numbers (due to increased activity of bone marrow)

Polycythemia

2. Secondary polycythemia

- Physiological cause
- Due to hypoxia secondary to:
 - High altitude
 - Heart failure
 - Respiratory failure

RBC Indices

- Help to realize the size, Hb-level, and type of RBC abnormalities
 - MCV → Mean Corpuscular Volume
 - MCH → Mean Corpuscular Hemoglobin
 - MCHC → Mean Corpuscular Hemoglobin Concentration
 - Color index

Parameter	Calculation	Male	Female
Hematocrit, or packed cell volume (<i>PCV</i>) (%)		47	42
Red cell count (<i>RCC</i>) (million/mm ³)		5.5	4.8
Hemoglobin concent. in blood [Hb] (g/dL)		16	14
Mean corpuscular volume (<i>MCV</i>) (μm ³)	$\frac{PCV \times 10}{RBCs \text{ count}}$	87	
Mean corpuscular hemoglobin (<i>MCH</i>) (pg/RBC)	$\frac{Hb \text{ conc.} \times 10}{RBCs \text{ count}}$	29	
Mean corpuscular hemoglobin concentration (<i>MCHC</i>) (g/dL)	$\frac{Hb \text{ conc.} \times 100}{PCV}$	34	
Color index	$\frac{Hb \text{ conc. (\% of normal)}}{PCV (\% \text{ of normal})}$	0.9 - 1.1	

RBC Indices

1. **MCV** → used to identify RBC according to their size

a. **Macrocytic**

- RBC that are bigger in size
- Pernicious or megaloblastic anemia, Vit. B-12 deficiency)

b. **Normocytic** → RBC with normal size ~ 7 UM

c. **Microcytic** → RBC that are smaller in size

RBC Indices

1. MCV:

- The average volume of a single RBC
- Expressed in cubic microns
- Normal = 90 (78-90)
- Macrocytic RBC – when MCV is increased
- Microcytic RBC – when MCV is decreased
- Normocytic RBC – when MCV is normal
- MCV is more in pernicious anemia and megaloblastic anemia in which RBCs are macrocytic in nature
- MCV is less in microcytic anemia

RBC Indices

2. **MCH** → used to identify Hb status

a. **Normochromic** → normal Hb synthesis → 33%/dl

b. **Hypochromic**

- < 33 g/100 ml
- Low values indicate deficient hemoglobin synthesis

c. **Hyperchromic**

- **High MCHC values do not occur**, because the Hb concentration is close to saturation point in red cells

RBC Indices

2. MCH:

- The amount of hemoglobin present in one RBC
- Expressed in picogram
- Normal = 30pg (27-32pg) – normochromic
- Decreases or remains normal in pernicious anemia and megaloblastic anemia
- Decreases in hypochromic anemia

RBC Indices

3. MCHC:

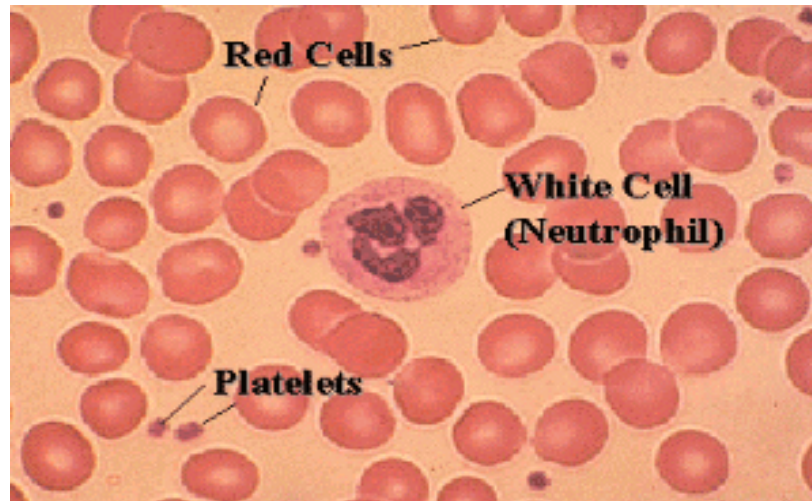
- Concentration of hemoglobin in one RBC
- The amount of hemoglobin expressed in relation to the volume of one RBC
- Expressed in percentage
- Normal = 30% (30-38)
- Decreased in iron deficiency anemia – RBCs are microcytic and hypochromic

RBC Indices

4. Color index (CI):

- The ratio between percentage of hemoglobin and percentage of RBCs in blood
- Average hemoglobin content in one cell of a patient compared the average hemoglobin content in one of a normal person
- Normal = 1.0 (0.8-1.2) – normocytic, normochromic
- Increased in pernicious and megaloblastic anemia
- Decreased in iron deficiency anemia

The Physiology of Blood



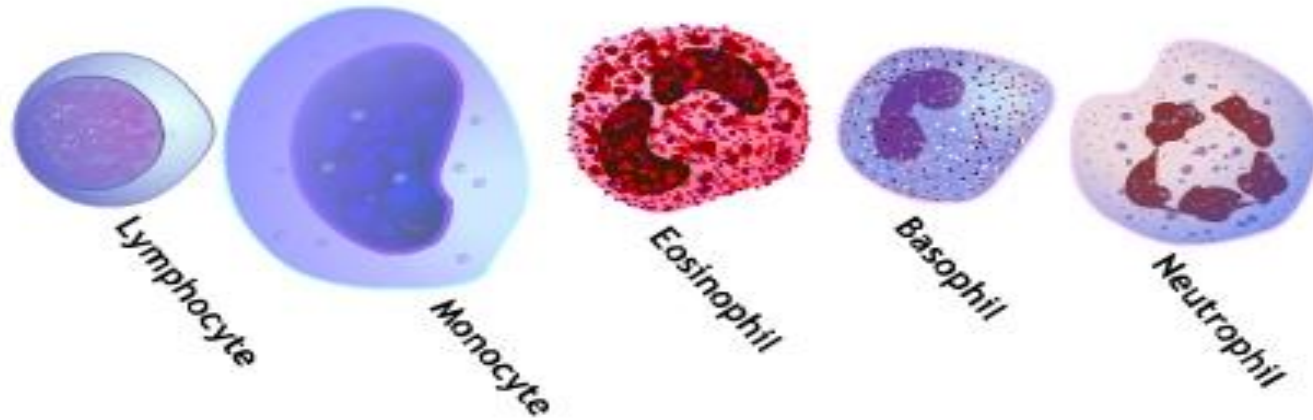
White Blood Cells (Leukocytes)

Leukocytes

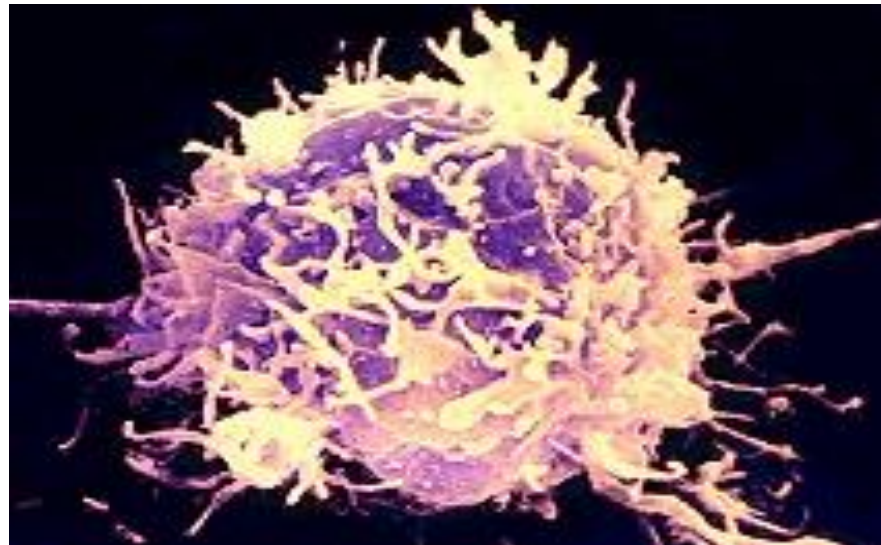
- **Characteristics:**
 - Constitute 1% of the total blood volume
 - Less numerous than RBCs
 - Normal WBC count: 4000 – 11000/mm³ (average 7000/mm³)
 - Are highly mobile and reach tissue fluids
 - **Function:** defense/protection against disease

Leukocytes

- Types:
- Granular leukocytes:
 - Neutrophils → 50-60%, 3000-7000/mm³
 - Eosinophils → 1-4%, 100-440/mm³
 - Basophils → 0.3-0.5%, 20-50/mm³
- Agranular leukocytes:
 - Lymphocytes → 20-40%, 1500-3500/mm³
 - Monocytes → 2-8%, 100-700/mm³ → circulate in blood
 - Monocytes change into macrophages in tissues
 - Macrophages live longer

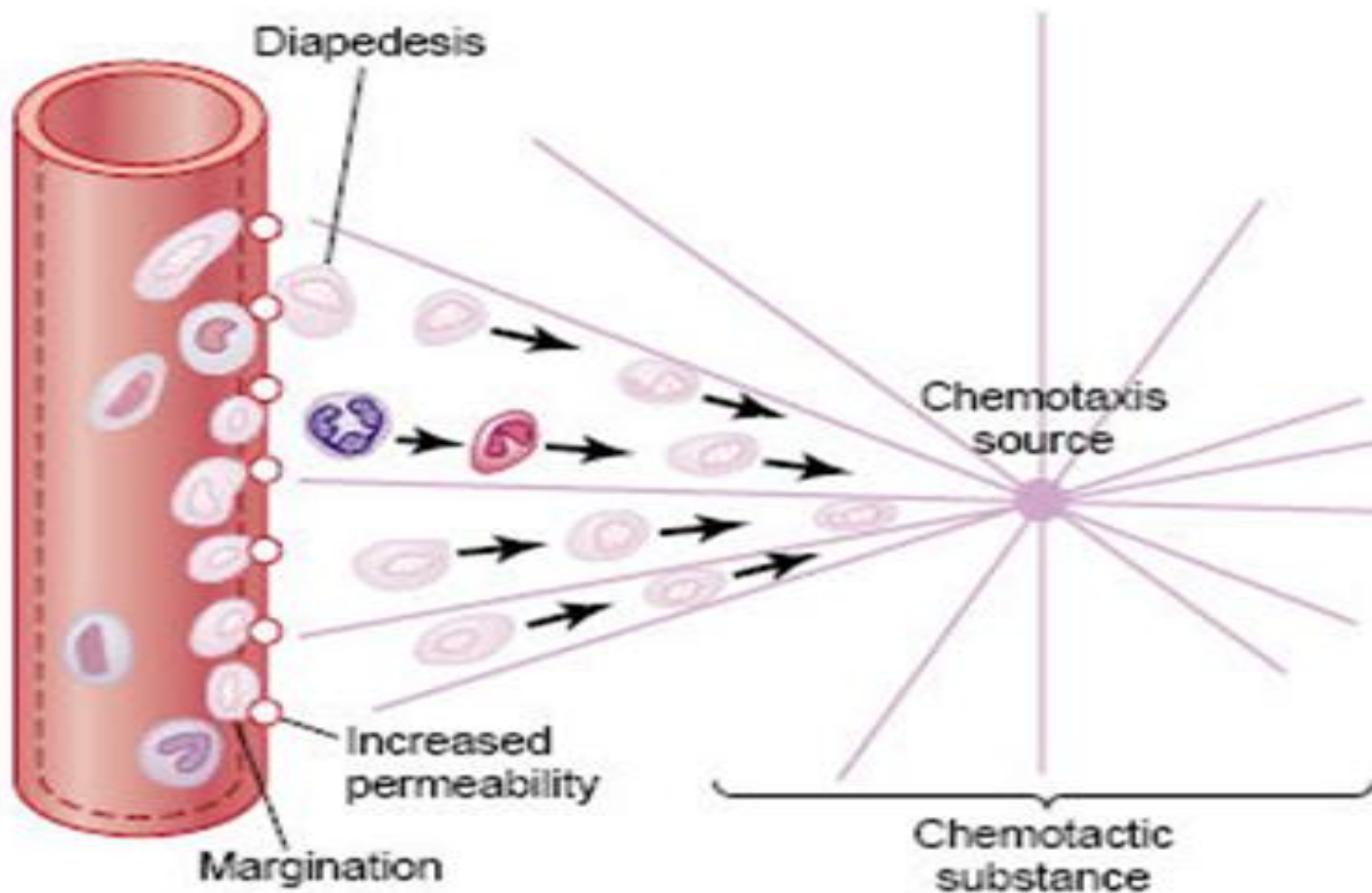


Macrophages



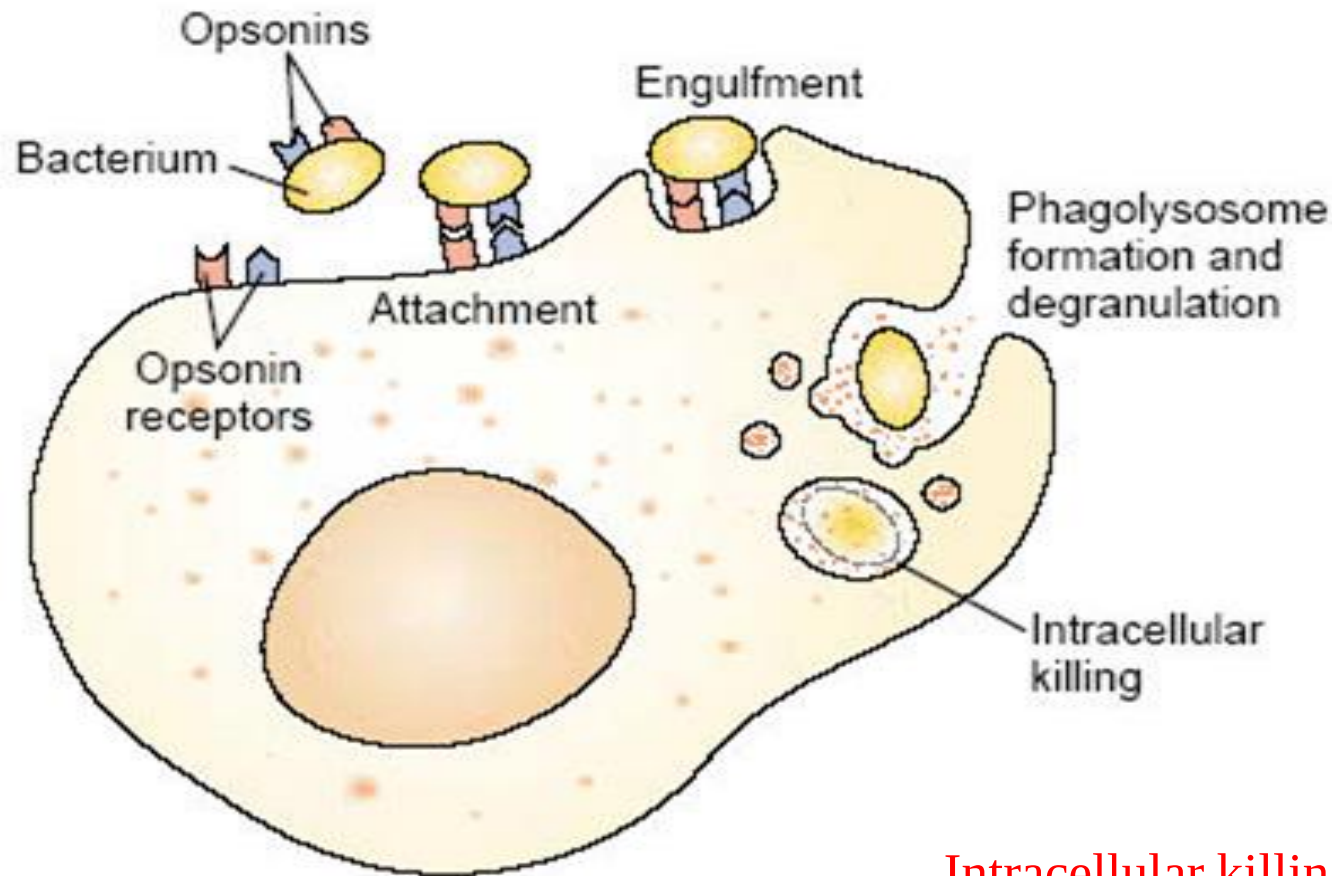
Leukocytes

- Mechanisms of mobility:



Leukocytes

- Mechanism of Phagocytosis:
 - Phagocytosis is initiated when a particle such as a bacterium, a dead cell, or tissue debris binds with receptors on the cell membrane phagocytes
 - Opsonization → Attachment → Engulfment → Intracellular killing
 - **Opsonin** → a protein fragment in the blood that binds to the surface of microbes and brought to phagocytes



Intracellular killing: by Lysosomes and Oxidizing agents produced from macrophages, like lipases, peroxisomes, H_2O_2 , OH^- , that are lethal for bacterial cell membranes

Functions of Leukocytes

- Neutrophils
 - The most abundant WBCs
 - The first line of defense
 - Phagocytic cells (ingest bacteria)
 - Their number increase during infection
- Basophils
 - Synthesize & store histamine, bradykinin, and serotonin
 - Release heparin → act as natural anti-clotting agent

Functions of Leukocytes

- Eosinophils
 - Fight parasitic infection and allergy
 - Absorb histamine during allergic conditions → anti-inflammatory effect
 - Produce hydrolytic enzymes to kill big parasites
- Monocytes
 - Highly phagocytic cells
 - They leave the circulation, enter to the tissue, and differentiate into macrophages
 - Macrophages are the largest WBCs

Tissue macrophage system (reticuloendothelial system)

Monocytes are formed in the bone marrow



Enter the circulation



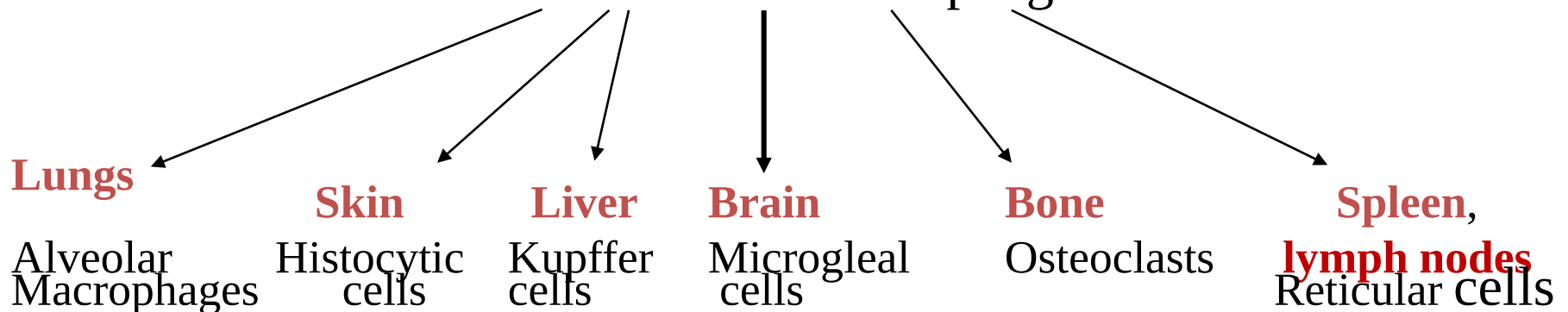
Leave the circulation, enter the tissue,



↑size, ↑lysosomal activities



Become tissue macrophages



Functions of Leukocytes

- **Lymphocytes:**
 - Responsible for Specific immunity:
 - Cellular immunity → T-lymphocyte mediated
 - Humoral immunity → B-lymphocyte mediated

B-Lymphocytes

- Formed in bone marrow and develop in liver or spleen
- As B cells mature during embryonic development, they develop **surface receptors** that allow them to **recognize specific antigens**
- Then they travel in the bloodstream, distributing throughout the lymph nodes, spleen, and tonsils
- Once B cells reach their destination, they remain inactive until they encounter a foreign cell with an antigen that matches their particular receptor site

B-Lymphocytes

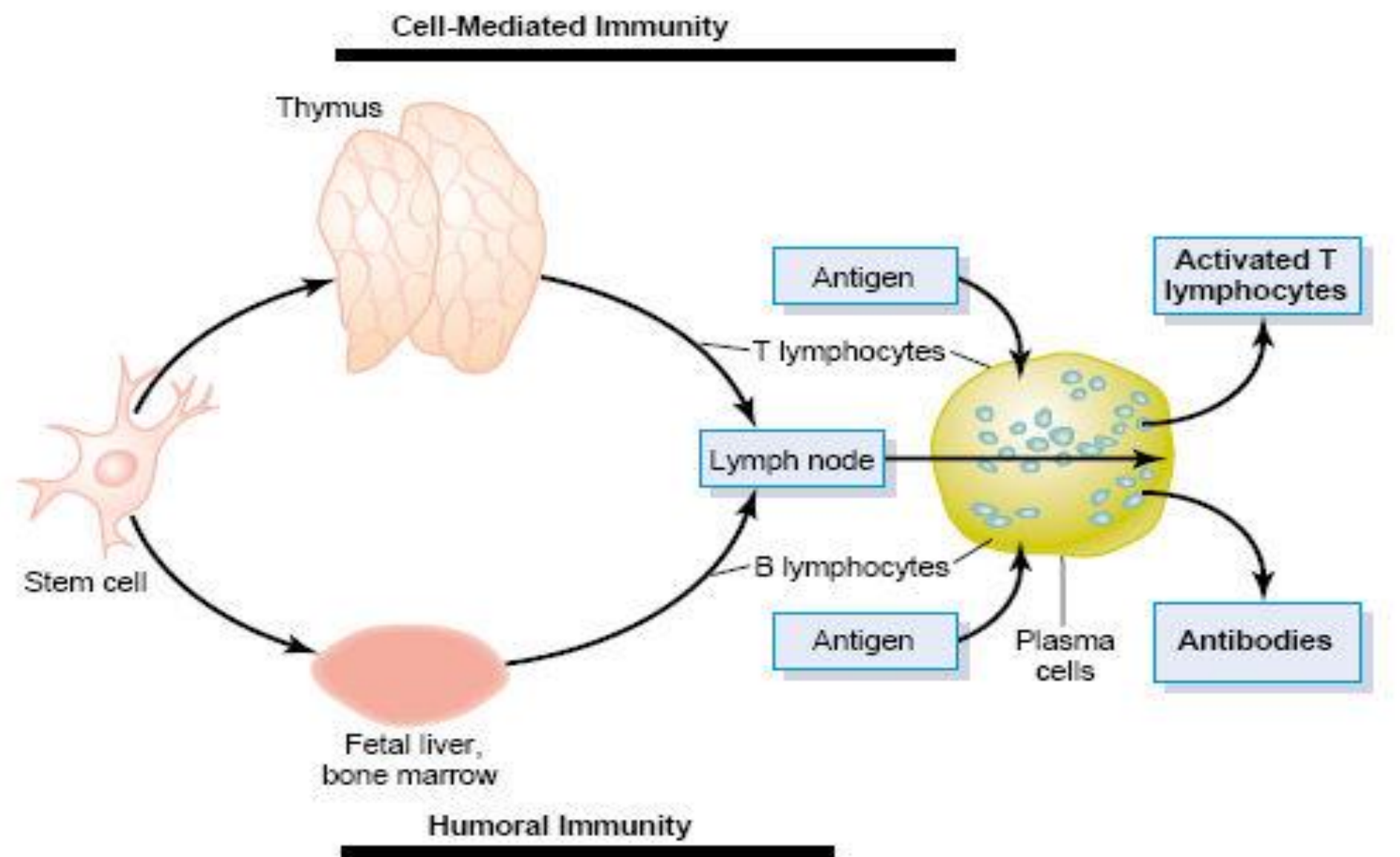
- Foreign antigens can be presented to B-cell directly or by **Antigen Presenting Cells** (macrophages and helper T)
- Encounter of an antigen causes differentiation of B-cells to plasma cells and memory cells
- **Plasma cells – secrete antibodies** which enter in to the blood stream through lymphatic circulation
- Different classes of antibodies (Immunoglobulin's (Ig):
 - **IgA + IgG + IgE + IgM + IgD**

B-Lymphocytes

- **IgA** → found in mucous membranes (RT, GIT, saliva, tears, breastmilk)
- **IgG** → most common type (75% of serum antibodies)
- **IgE** → defend allergen reaction
- **IgM** → the first antibody to appear in response to initial exposure to an antigen (+ the largest antibody)
- **IgD** → surface receptor (stimulates B-cells to mature into plasma cells)
→ Mainly present in the URT

B-Lymphocytes

- **Memory B-cells**
 - Record the information about the foreign antigen so antibodies can be made more quickly, and in greater amount, in case of a second exposure
 - The second response is much faster and stronger than the first response
 - They are long lived cells



T-Lymphocytes

- Produced in the bone marrow and matured in the Thymus gland
- Macrophages phagocytize invading microbes and present parts of the microbe (antigens) to T-lymphocytes and stimulates T-cell differentiation
- Types of T-cells:
 - Cytotoxic T- cells (Tc)
 - Helper T- cells (Th)
 - Suppressor T- cells (Ts)
 - Memory T- cells (Tm)

Leukocyte Disorders

1. Leukemia

- Increased/cancerous production of WBCs
- The WBCs produced, though numerous, are immature and non functional

- Effects of leukemia

- a. Metabolic starvation

- The increased WBC consume too much metabolic substrates
 - Thus energy source is depleted and too much use of amino acids by cancerous cells causes rapid deterioration of other body tissues

Leukocyte Disorders

- Effects of leukemia

- B. Anemia and increased bleeding tendency

- Increased production of WBCs takes the place of platelets & RBCs causing anemia + impaired blood clotting

2. Leukopenia

- Decreased production of WBCs
- May be due to bone marrow destruction by Drugs, poisons, X-rays...

The Physiology of Blood

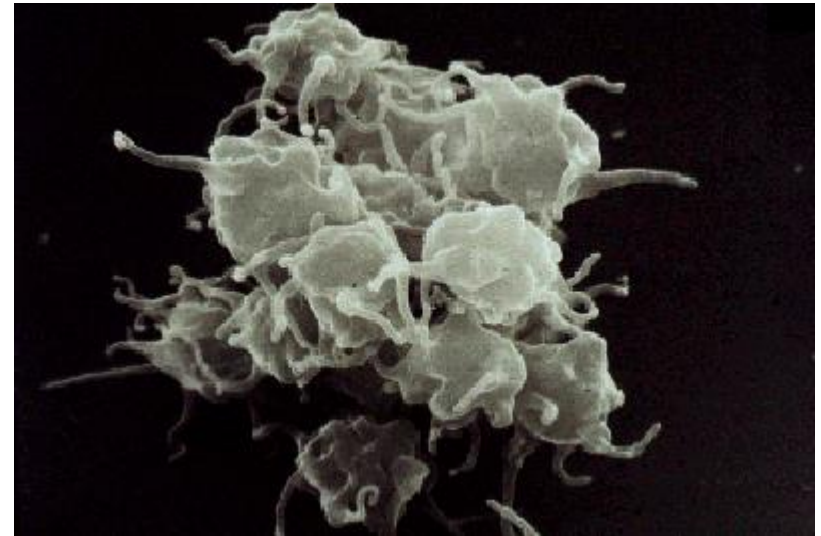
Platelets and Hemostasis

Platelets

- Small fragments that emerge from Megakaryocytes in red bone marrow (2-4 microns in diameter)
- Normal platelet count = 150,000 – 300,000/mm³
- Function: **hemostasis (clotting)**



Inactive platelets



Active platelets

Platelets

- Properties:

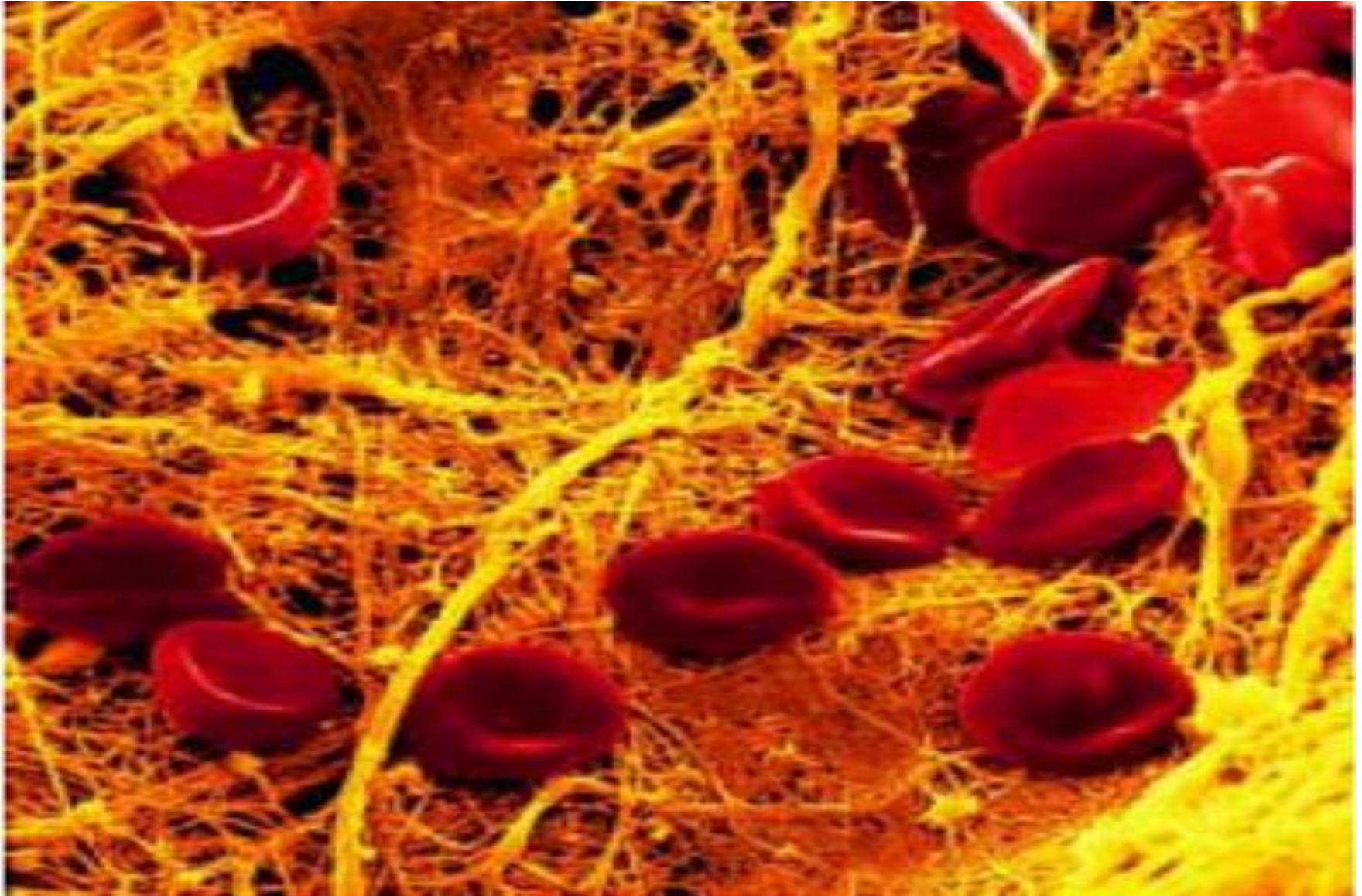
A. Adhesiveness :

- Platelets stick when they come in contact with wet and rough surfaces.
- **Glycoprotein's** in their cell membrane repel platelets not to adhere to the endothelium in normal conditions

B. Aggregation:

- When platelets are activated, they usually group together.
- Their aggregation and stickiness is mainly due to **ADP** and **Thromboxane A2** found in their cytoplasm

C. Agglutination → Platelets clump together and form clots



Hemostasis

- Prevention of bleeding
- Platelets are responsible for clotting
- Platelets synthesize chemicals which enhance clotting:
 - Thromboxane A₂
 - Serotonin
 - Histamine
 - Von Willebrand's factor (important for adherence of platelets)
 - Minerals like Cu²⁺ and Fe²⁺ that are involved in clotting processes

Hemostasis

- Stages:

1. Vascular spasm (vasoconstriction) → immediate myogenic response to injury
2. Platelet plug formation
3. Blood Clot formation (coagulation)
4. Fibrin mesh development (clot retraction)
5. Clot dissolution

Hemostasis

1. Vascular spasm:

- Small blood vessels and arterioles constrict at local injured regions and decrease blood flow to injured areas.

- General mechanism of vasoconstriction involves:

a. **Local myogenic spasm** → contraction of the injured smooth muscle

b. **Local metabolites:**

- Vasoconstrictor substances released from platelets and traumatized tissues → serotonin + histamine, TXA2 + ...

Hemostasis

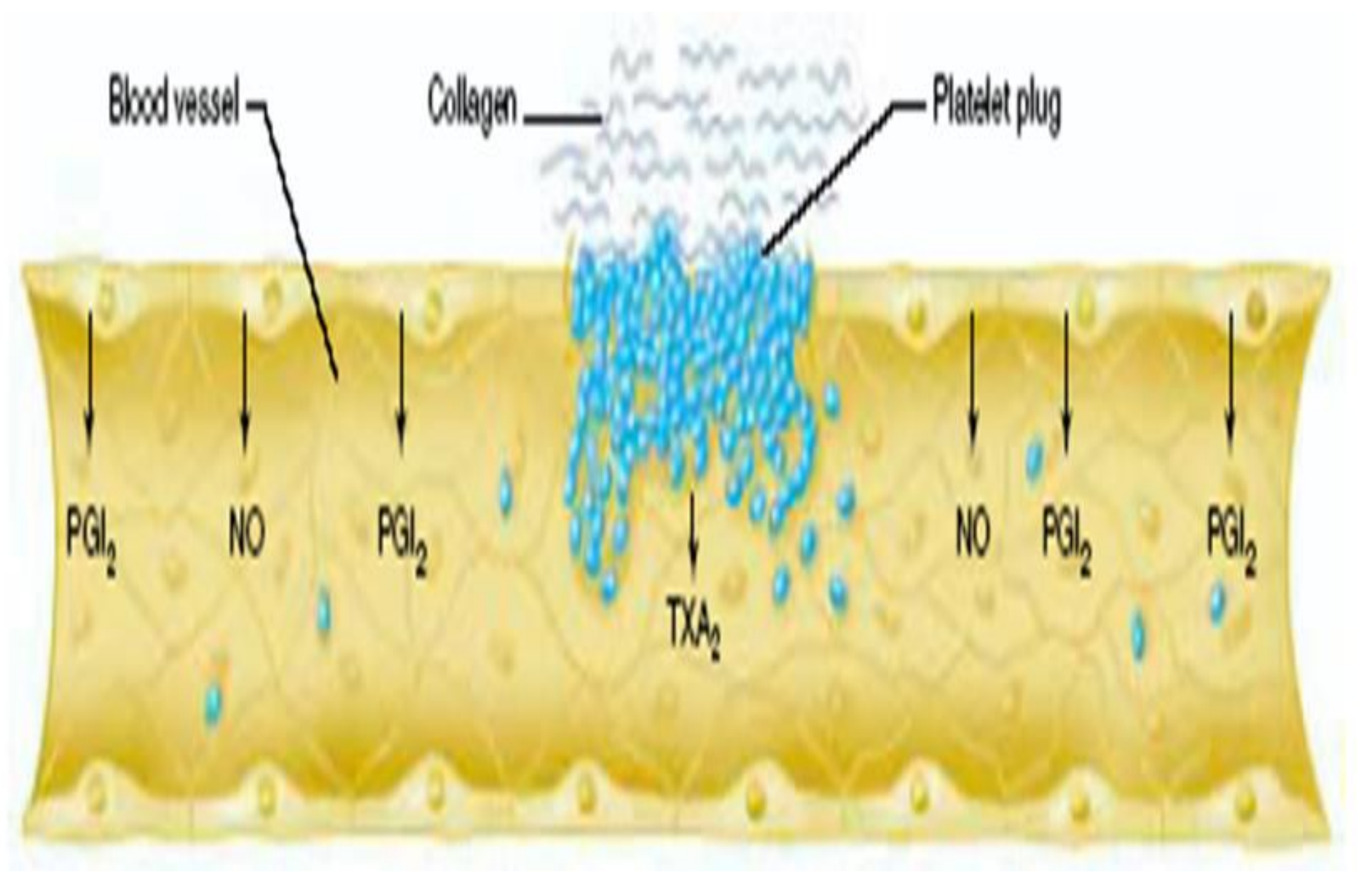
1. Vascular spasm:

c. Nervous reflexes:

- Reflex sympathetic discharge causes constriction of the traumatized blood vessels

- **NB**

- The spasm can last several minutes to hours



Hemostasis

2. Platelet plug formation:

- Normally, platelets do not stick to each other or to the endothelial lining of blood vessels
- Upon damage to blood vessel endothelium (which exposes collagen) platelets:
 - Adhere to collagen
 - Stick to exposed collagen fibers and form a platelet plug
 - Release **serotonin** and **ADP**, which attract additional platelets
- The platelet plug is limited to the immediate area of injury

Hemostasis

3. Blood clot formation (coagulation):

- A phase where blood loses its fluidity and becomes a jelly-solid mass
- Slow but long-lasting effect to stop bleeding
- Coagulation undergoes three steps of reactions:
 - Formation of **prothrombin activator** through
 - **Intrinsic pathway**
 - **Extrinsic pathways**
 - Conversion of **prothrombin into thrombin** by the action of prothrombin activator (**prothrombinase**)
 - Conversion of **fibrinogen into fibrin** thread by the action of thrombin

Hemostasis

- Blood clotting factors:
 - They are produced by:
 - Liver
 - Platelets
 - Endothelial cells

1. Factor-I: **Fibrinogen**
2. Factor-II: **Prothrombin**, α_1 - globulin
3. Factor-III: **Tissue factor**, tissue thromboplastin, a phospholipoprotein
4. Factor-IV: **Calcium ion**, essential for clotting
5. Factor-V: **Labile factor**, β -globulin. Deficiency leads to parahemophilia
6. Factor-VII: **Stable factor**, α_1 -globulin
7. Factor -VIII: **Antihemophilic factor-A**, β_2 -globulin
8. Factor -IX: **Christmas factor**, AHF-B, α_1 - globulin
9. Factor -X: **Stauter-power factor**
10. Factor-XI: **Antihemophilic factor-C**, plasma thromboplastin antecedent γ -globulin
11. Factor-XII: **Contact factor**, Hagen factor, glass factor
12. Factor-XIII: **Fibrin stabilizing factor**, β -globulin, polymerizes fibrin threads

Clotting Factors in Blood and Their Synonyms

Clotting Factor	Synonyms
Fibrinogen	Factor I
Prothrombin	Factor II
Tissue factor	Factor III; tissue thromboplastin
Calcium	Factor IV
Factor V	Proaccelerin; labile factor; Ac-globulin (Ac-G)
Factor VII	Serum prothrombin conversion accelerator (SPCA); proconvertin; stable factor
Factor VIII	Antihemophilic factor (AHF); antihemophilic globulin (AHG); antihemophilic factor A
Factor IX	Plasma thromboplastin component (PTC); Christmas factor; antihemophilic factor B
Factor X	Stuart factor; Stuart-Prower factor
Factor XI	Plasma thromboplastin antecedent (PTA); antihemophilic factor C
Factor XII	Hageman factor
Factor XIII	Fibrin-stabilizing factor
Prekallikrein	Fletcher factor
High-molecular-weight kininogen	Fitzgerald factor; HMWK (high-molecular-weight) kininogen
Platelets	

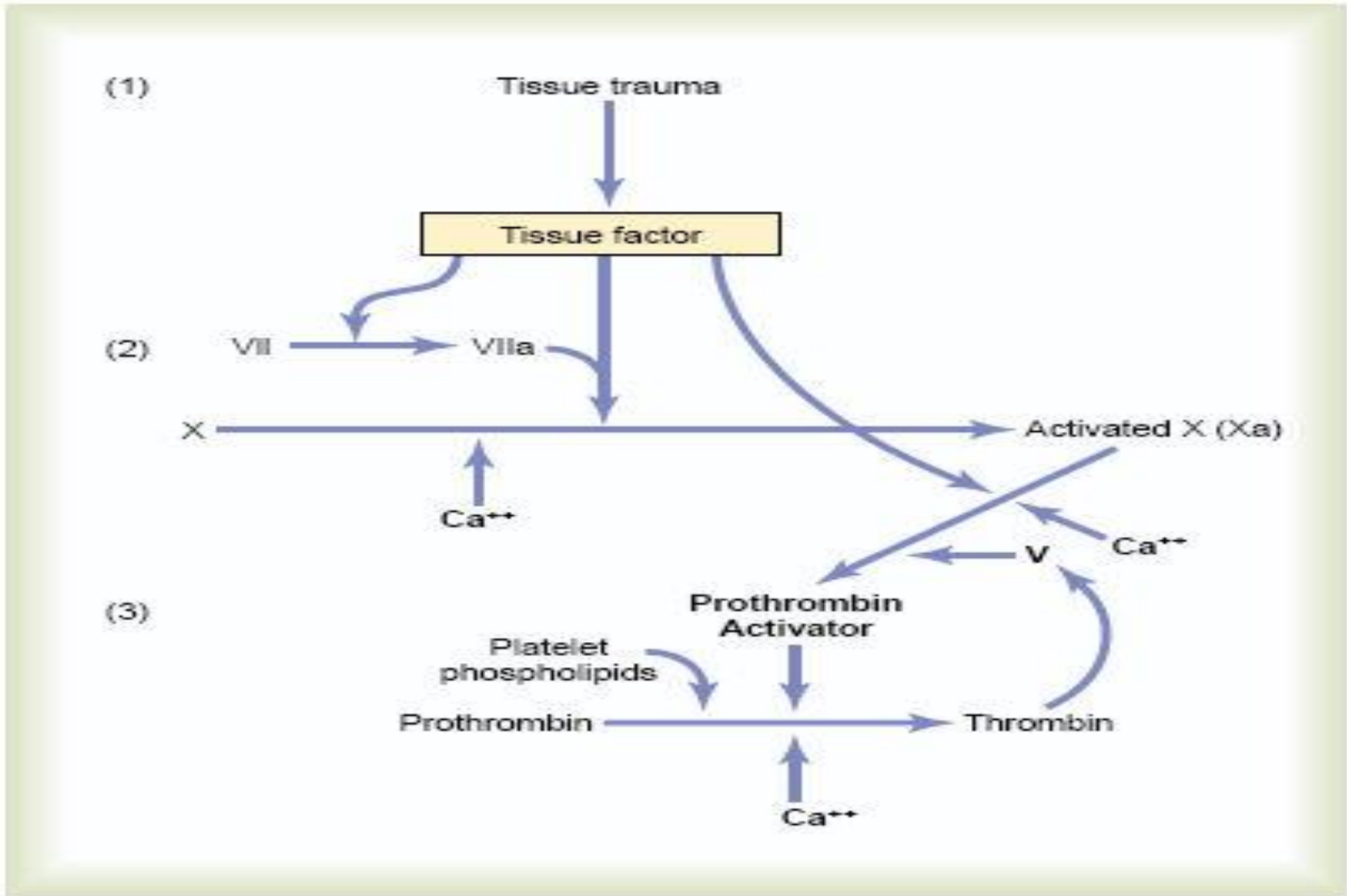
Hemostasis

- Extrinsic pathway
 - Begins with **traumatized tissues** that release tissue thromboplastin (tissue factor)
 - Tissue factor together with factor VII activates factor X to Xa
 - Activated Factor Xa with Ca^{2+} and Factor V forms a Prothrombin activator that in turn activates and converts:

Prothrombin $\rightarrow$ Thrombin

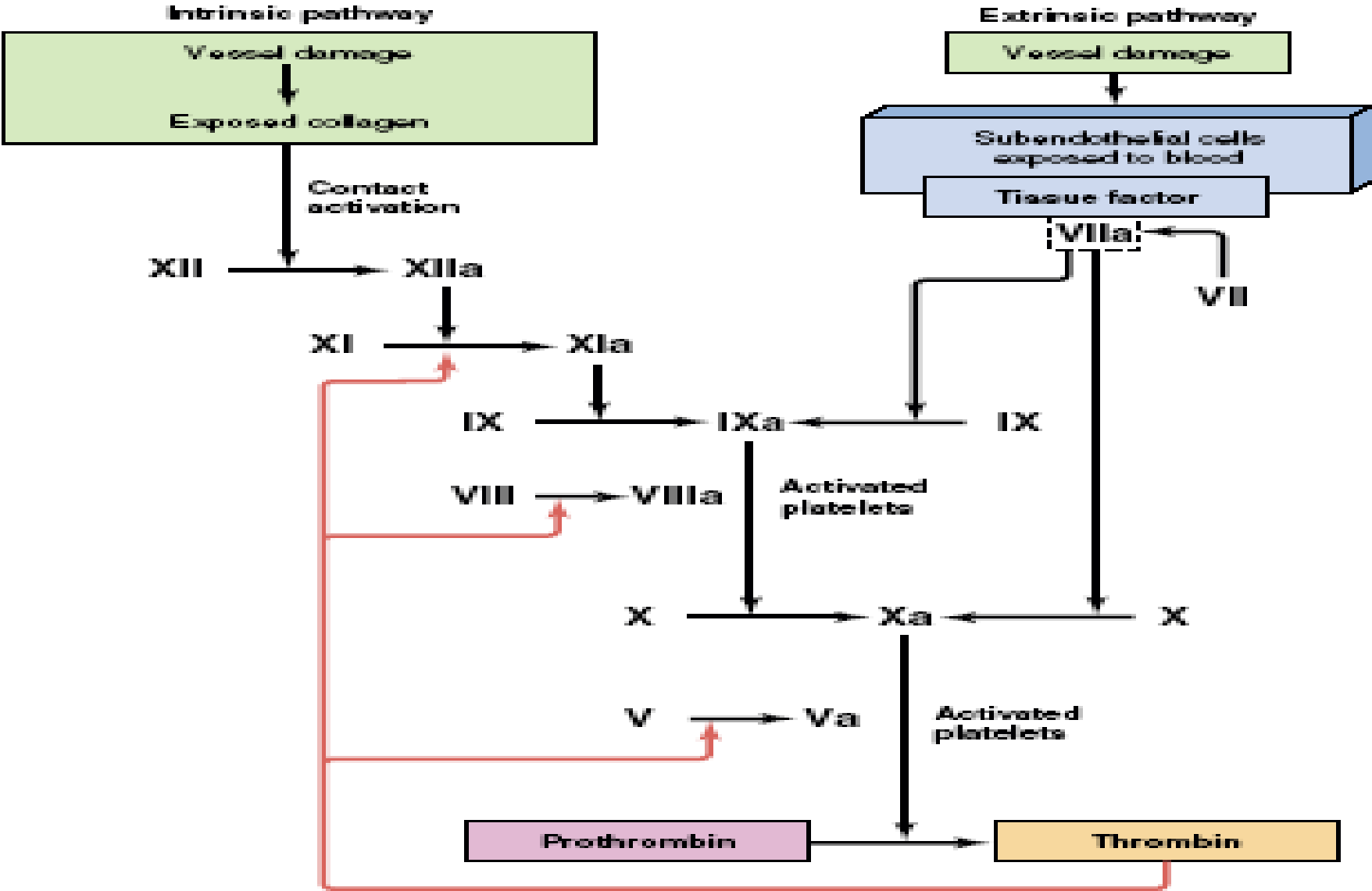
- Thrombin together with Ca^{2+} converts:

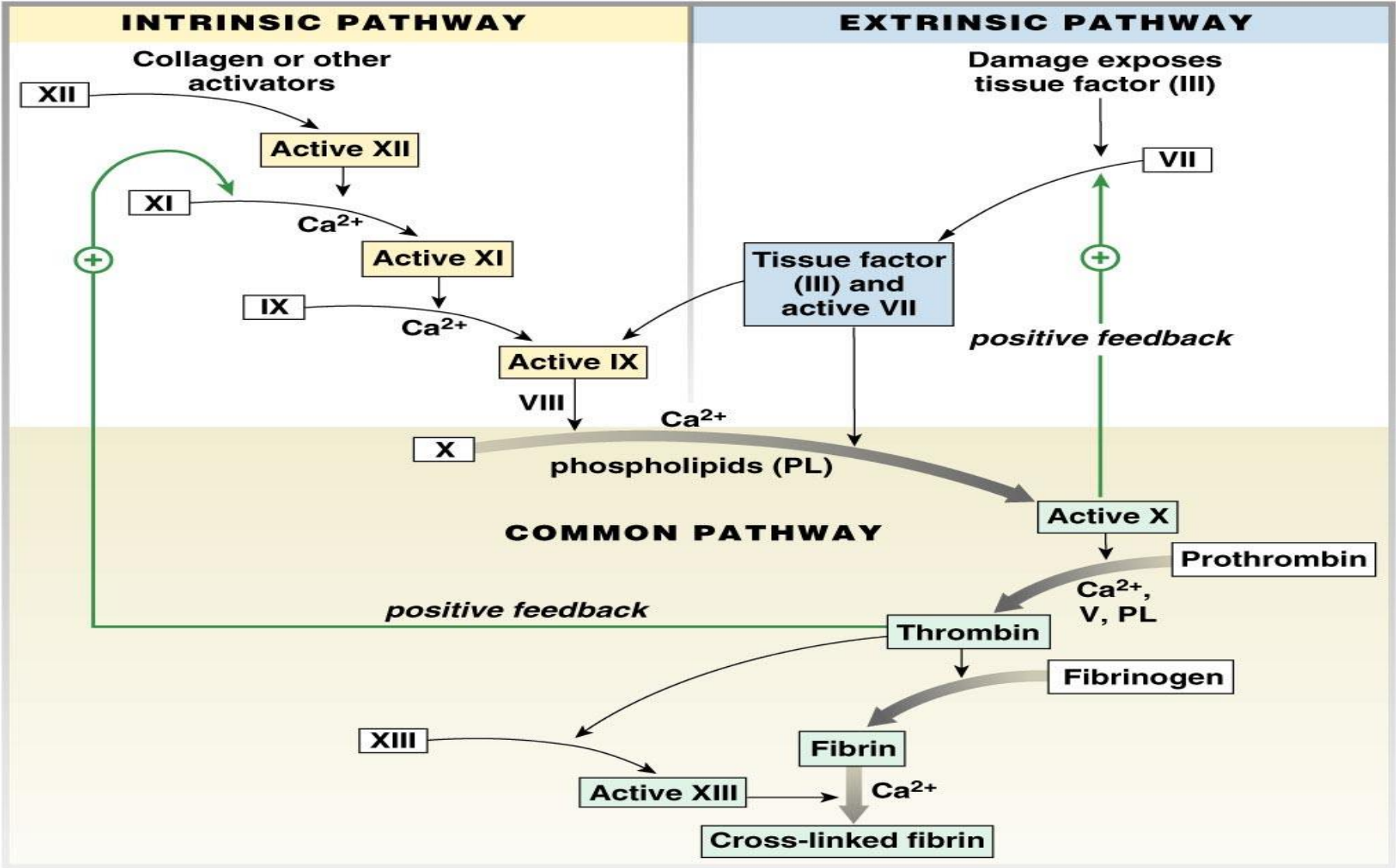
Fibrinogen $\rightarrow$ Fibrin (loose solid clot)



Hemostasis

- Intrinsic pathway
 - Begins with trauma to the blood itself or exposure of the blood to collagen at injured blood vessel wall
 - The blood then activates factor XII to its active form XIIa
 - Then cascade of reactions continues until the formation of activated Factor Xa
 - Activated Xa combines with Ca^{2+} and thrombin to produce the Prothrombin activator (common pathway)

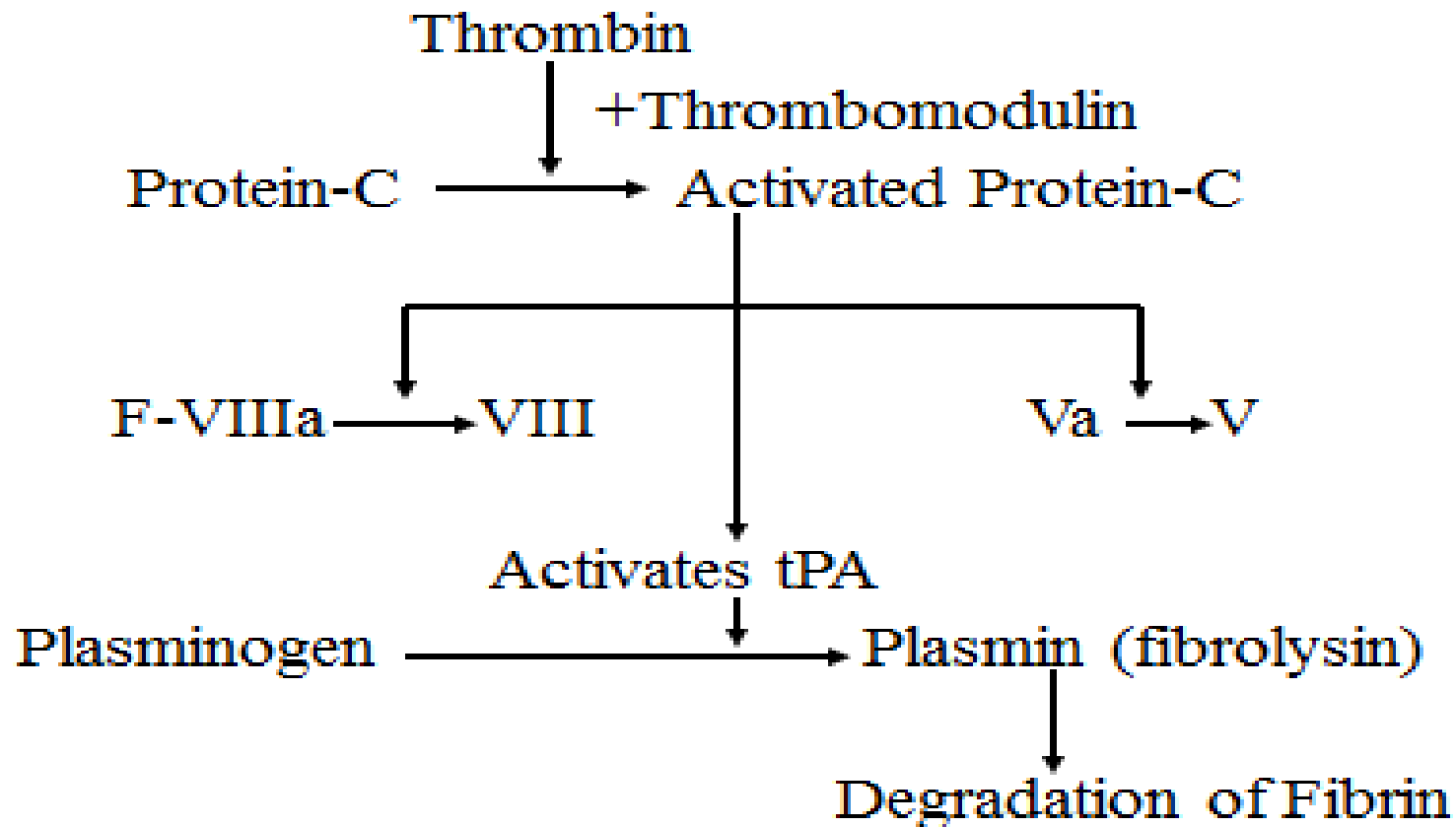




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Fibrinolysis: dissolution of clot

Steps:



Bleeding Disorders

- Excessive bleeding can be caused due to 3-major factors:

1. Thrombocytopenia (platelet deficiency)

- Bleeding at small vessels, capillaries and Venules
- Platelets usually plug small blood vessels, so their absence causes **thrombocytopenic purpura** (small purplish blotches)

2. Vitamin-k deficiency

- Vitamin K (found in green vegetables) is necessary for the synthesis of Prothrombin, factors: II, VII, IX and X
- Adequate Vit-K is usually formed by bacteria's in the large intestine

Bleeding Disorders

2. Vitamin-k deficiency

- Deficiency of Vit-K can cause hemorrhagic disorder
- Vit-K deficiency occurs in case of obstructive jaundice, chronic diarrhea, liver disease (hepatitis, cirrhosis, malignancy)
- Vit-K requires fat for absorption into the blood stream
- Deficiency of fat-absorption (or deficiency of bile formation from the liver) can cause lack of Vit-K

Bleeding Disorders

3. Hemophilia

- An X-linked inherited recessive bleeding disorder
- Occur almost exclusively in men
- Bleeding occurs at big blood vessels

a. Hemophilia-A (classic hemophilia):

- Due to inadequate production of Factor-VIII

b. Hemophilia-B (charismas disease):

- Due to inadequate production of Factor-IX

- When Factor-VIII and/or IX are absent, Factor-X can not be activated
- Factor-X is the main enzyme that converts Fibrinogen to Fibrin

Coagulants and Anticoagulants

- There are more than 50 different types of anticoagulants and pro-coagulants circulating in the blood
- In normal circumstances **Anticoagulants** dominate over **procoagulants**
- During injury, procoagulants dominate near the site of injury
- **Pro-coagulants:**
 - Thrombin
 - Snake venom
 - Thromboplastin...

Coagulants and Anticoagulants

- Anticoagulants:

1. **Heparin** → anti-thrombin and inactivates factor IX, X, XII

2. **Warfarin and dicumarol:**

- Inhibit Vit-K that is necessary to produce factor X, IX and VII

3. **EDTA** (Ethylene diamine tetra acetic acid):

- Removes Ca^{2+} from blood

4. **Oxalate**

- Combines with Ca^{2+} and forms Ca^{2+} oxalate precipitate

5. **Citrates** → form complex with Ca^{2+} and decreases its level in plasma (in vitro for a blood bank)

Bleeding Tests

1. Bleeding time

- The time interval between the start of bleeding (oozing) until arrest of blood
- Normal duration → 3-6 min

2. Clotting time:

- The duration of time for the blood to clot (normal duration is 3-8 min)

3. Prothrombin time:

- The duration of formation of Prothrombin after addition of oxalate and Ca^{2+} ions to the blood

Thromboembolic Formation

- **Thrombus:**
 - A clot that develops and persists in blood vessels
 - Thrombi can block circulation, resulting in tissue death and organ failure
 - Coronary thrombosis – thrombus in blood vessel of the heart
 - Myocardial infarction (heart attack)

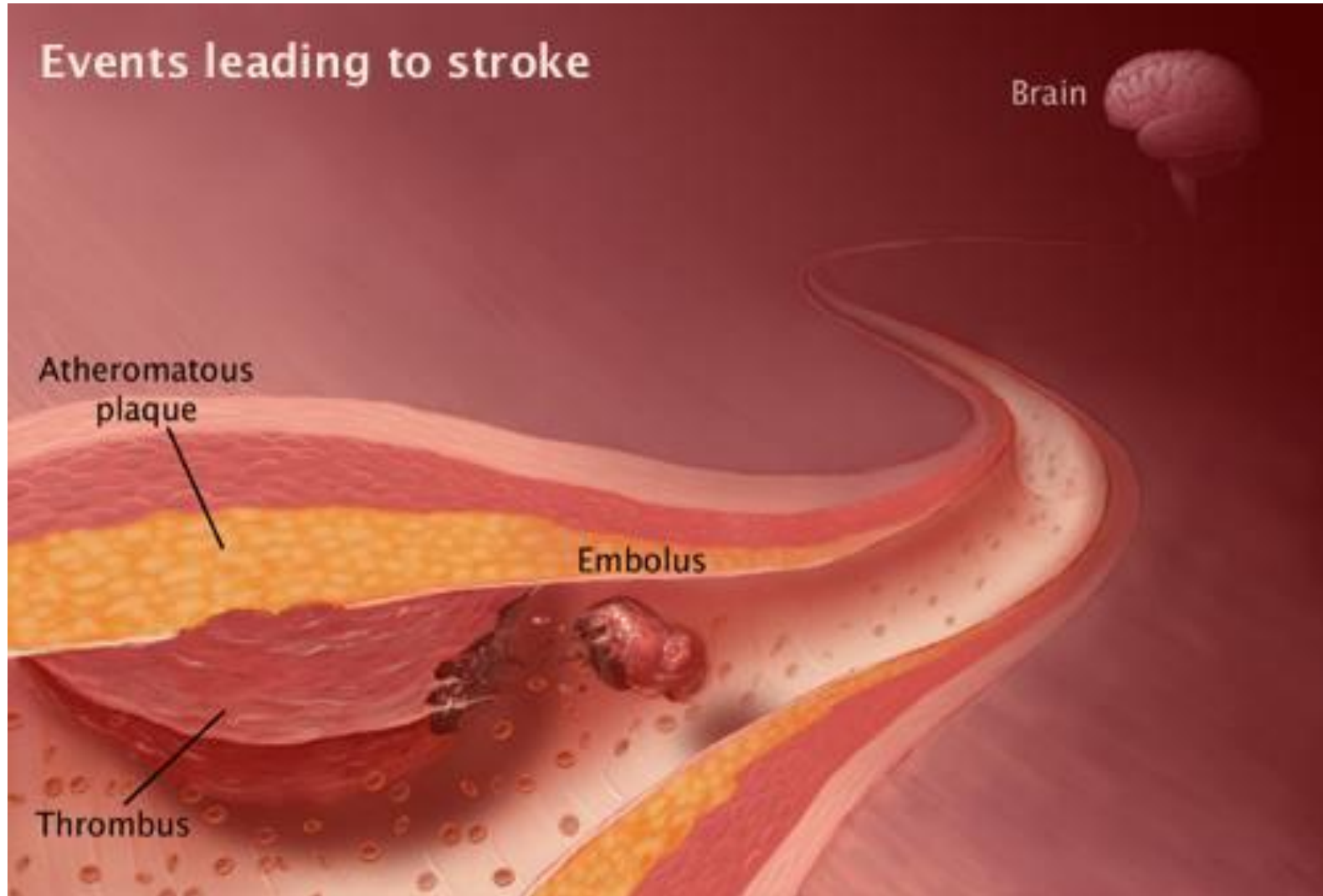
Thromboembolic Formation

- **Embolus**
 - A thrombus (blood clot) freely floating in the blood
 - The thrombi detaches, forced by flowing blood and cause the clot to flow through the circulation.
 - Pulmonary emboli can impair the ability of the body to obtain oxygen
 - Cerebral emboli can cause strokes

Thromboembolic Formation

- Causes of thrombo-embolism
 - Damage to the endothelium (Atherosclerosis) – Rough endothelial surfaces of a vessel
 - Sluggishness of blood flow
 - Increase in viscosity of blood due to high platelet count or high RBC count

Thromboembolic Formation



The Physiology of Blood

Blood Groups

Blood Groups

- Blood grouping depends on the presence of antigens (agglutinogens) on the surface of RBCs
- RBCs contain protein/glycoproteins/lipoproteins on their surface with antigenic effect
- These antigens are:
 - Unique to the individual
 - Recognized as foreign if transfused into another individual who does not have those same antigens on RBC membrane
- Presence or absence of these antigens is used to classify blood groups

Blood Groups

- Major blood groups:
 1. ABO blood group
 2. Rh blood group

ABO Blood Group

- Based on the $\pm$ of two antigens (A and B), ABO blood groups can be classified into four groups:
 1. When antigen-A is present on RBCs $\rightarrow$ blood type-A
 2. When antigen-B is present on RBCs $\rightarrow$ blood type-B
 3. When both antigen-A & B exist together $\rightarrow$ blood type-AB
 4. When neither antigen-A nor B are present $\rightarrow$ blood type-O

ABO Blood Group

Blood type	Antigen	Antibody	Genotype
A	A	anti-B	AO, AA
B	B	anti-A	BO, BB
AB	A & B	Neither	AB
O	neither A nor B	anti-A & anti-B	OO

Relative Frequencies of the Different Blood Types.

The prevalence of the different blood types among one group of persons studied was approximately:

O	47%
A	41%
B	9%
AB	3%

It is obvious from these percentages that the O and A genes occur frequently, whereas the B gene is infrequent.

Antibodies

- In the ABO blood grouping, plasma contains two genetically determined agglutinins (antibodies):
 - Anti-A antibodies (agglutinin- α)
 - Anti-B antibodies (agglutinin- β)

Antibodies

- Existence of agglutinogens and agglutinins
 1. When agglutinogen-A is present on RBCs → anti-B ab
 2. When agglutinogen-B is present on RBCs → anti-A ab
 3. When agglutinogen-A & B exist together → neither anti-A nor anti-B abs are present in the serum
 4. When no agglutinogen on RBCs → both anti-A and anti-B abs are present in the serum

ABO Blood Typing

- To determine a blood type, a drop of blood sample is mixed with a known prepared antibodies

Blood samples	Known agglutinins		Blood type
	Anti-A antibody	Anti-B antibody	
1	+	—	A
2	—	+	B
3	+	+	AB
4	—	—	O

(+) = Agglutination reaction

(—) = No agglutination reaction

ABO Blood Cross Matching

- To determine a blood type, a drop of blood sample is mixed with a known prepared antibodies

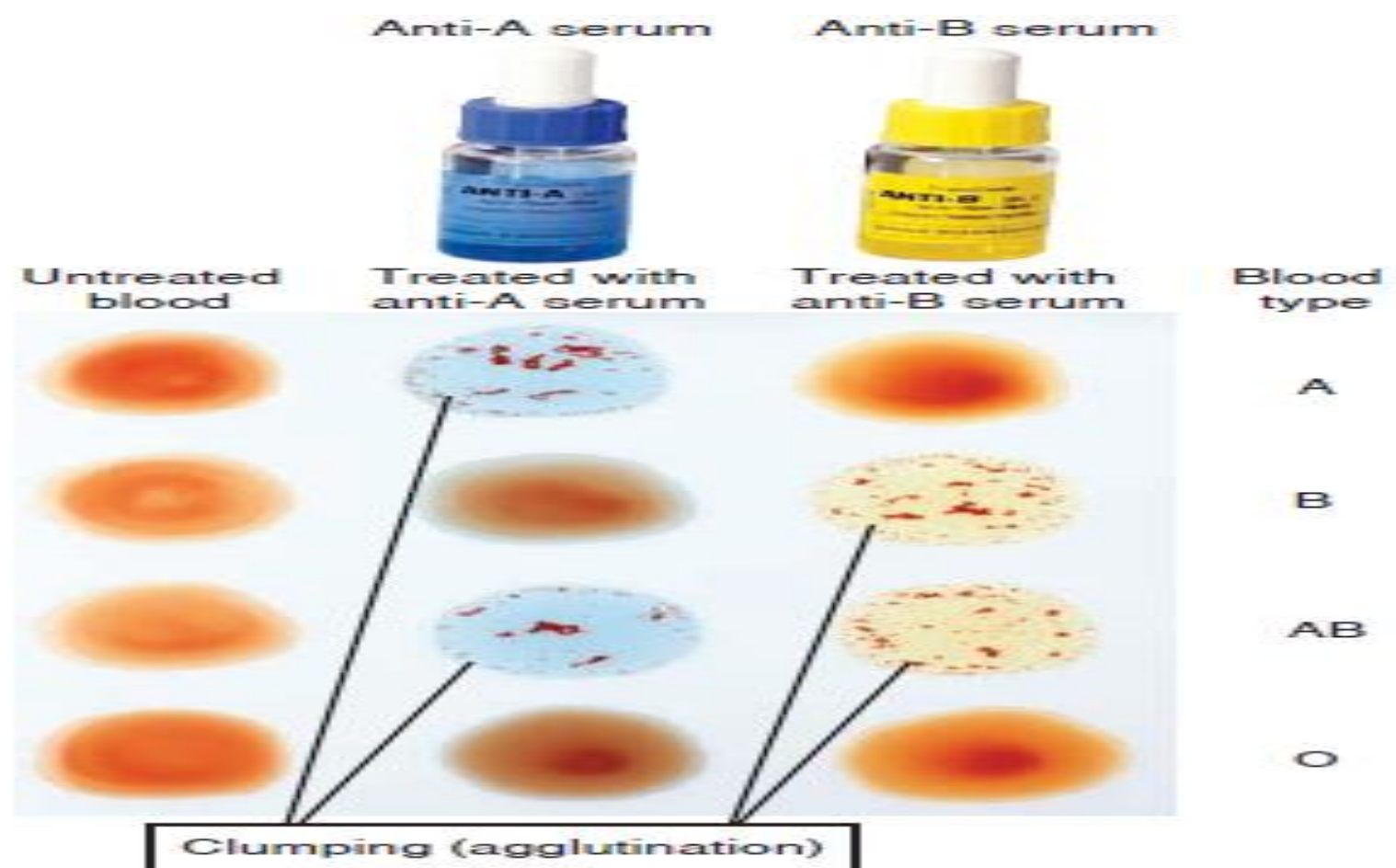
Donors Blood types	Recipients blood types			
	A	B	AB	O
A	—	+	—	+
B	+	—	—	+
AB	+	+	—	+
O	—	—	—	—

(+) = Agglutination reaction

(—) = No agglutination reaction

Blood Typing, Showing Agglutination of Cells of the Different Blood Types with Anti-A or Anti-B Agglutinins in the Sera

Red Blood Cell Types	Sera	
	<i>Anti-A</i>	<i>Anti-B</i>
O	-	-
A	+	-
B	-	+
AB	+	+



Blood type	A	B	AB	O
Compatible donor blood types (no reaction)	A, O	B, O	A, B, AB, O	O
Incompatible donor blood types (reaction)	B, AB	A, AB	—	A, B, AB

ABO Blood Cross Matching

- Blood group “O” is a universal donor
 - People with this blood types have no antigens on their cell-membrane surfaces and therefore can not agglutinate if transfused to any blood types
 - Even though they lack antigens, they have Anti-A & anti-B antibodies in their plasma
 - So, they can receive blood from persons with blood group “O” only

ABO Blood Cross Matching

- Blood group AB is a universal recipient
 - People with AB-blood group can take blood from any blood types, because they have no antibodies in their blood to cause agglutination reactions
 - AB can donate blood only to a person with blood AB, not to the others

Rh Blood Group

- It was first worked out on Rhesus monkey to the name Rh-system
- Though there are many kinds of Rh-antigens (C, D, E, c, d, e), the most potent one is antigen D.
- Grouped as Rh⁺ and Rh⁻ based on the presence or absence of agglutinin D on the surface of RBCs
- The presence of agglutinin-D on RBCs: Rh⁺
- Missing agglutinin-D RBCs: Rh⁻
- Anti-D antibodies are not normally present in the serum, produced secondary to exposure of an Rh⁻ blood to Rh⁺ blood (antigen-D)

Rh Blood Group

- In comparison to ABO blood group, the Rh-factor is slow acting and is not found in other secretion like saliva
- Rh Incompatibility → Erythroblastosis Fetalis
 - Rh+ father X Rh- mother = Rh + fetus
 - During birth, Rh+ antigens of the fetus enter to mothers blood through placenta and sensitizes her
 - Mother's blood produces anti-Rh antibodies (anti-D antibodies) against the Rh+ blood

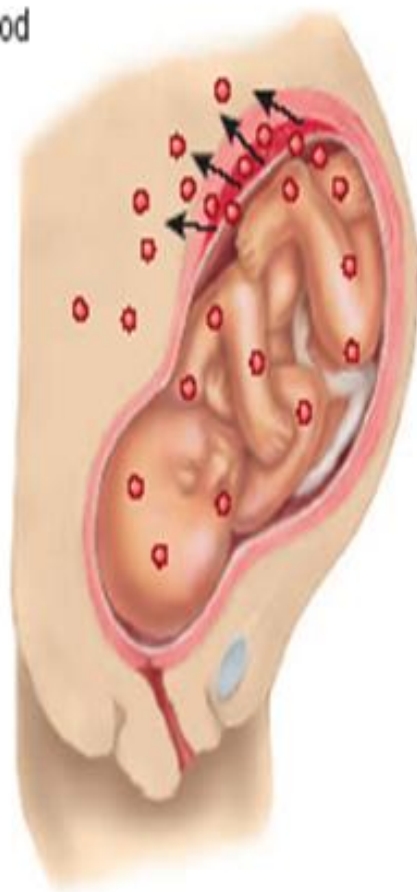
Rh Blood Group

- During the 2nd pregnancy and there after, Anti-Rh+ antibodies enter into the fetus and agglutinate (hemolyze) RBC's of fetus
- If the baby is born alive from the incidence, then there is a higher risk of being **anemic and jaundiced**
- The passage of the bile pigment to the brain causes **Kernicterus** (damage of the basal ganglia with subsequent motor disturbances)
- **Treatment**
 - **Rho Gam** injection after 1st birth which blocks mother from forming anti-Rh antibody
 - Limits risk for subsequent births

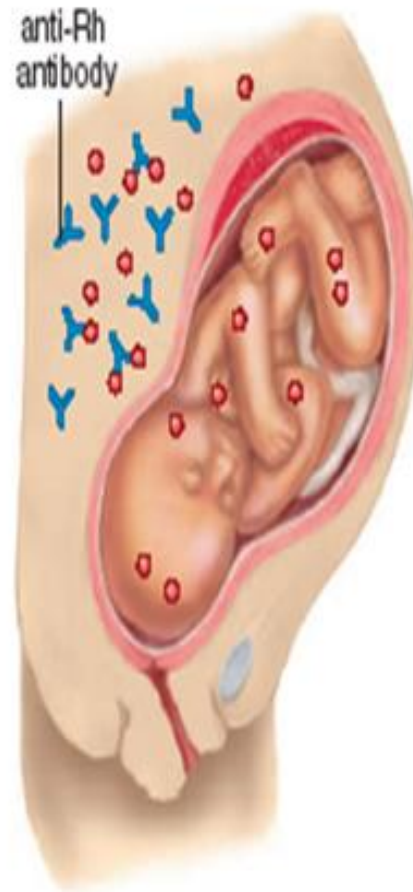
Hemolytic disease of the new born



Child is Rh positive;
mother is Rh negative.



Red blood cells leak
across placenta.



Mother makes anti-Rh
antibodies.



Antibodies attack Rh-positive
red blood cells in child.

Prevalence of ABO and Rh blood groups among ethnic groups

Races	Frequency of blood groups (%)				
	O	A	B	AB	Rh ⁺
Whites	45	41	10	4	85
Blacks	48	27	21	4	88
Chines	36	28	23	13	100
Ethiopians	47	28	20	5	97

Blood Transfusion

- May be defined as giving of a blood donor to recipient.
- May be whole blood transfusion or Packed red cells transfusion
- Whole blood transfusions are used:
 - When blood loss is substantial
 - In treating thrombocytopenia
- Packed red cells (cells with plasma removed) are used to treat anemia

Blood Transfusion

- **Transfusion Reactions**

- Transfusion reactions occur when mismatched blood is infused
- Donor blood cells are attacked by the recipient's plasma
 - Diminished oxygen-carrying capacity
 - Clumped cells that impede blood flow
 - Ruptured RBCs that release free hemoglobin into the bloodstream
 - Circulating hemoglobin precipitates in the kidneys and causes renal failure

Human Leukocyte Antigens (HLA)

- MHC (Major Histo-compatibility Complex)
- Genes that determine self from non-self antigens
- They are part of WBC's (particularly, T-cells)
- So antigens that cause graft rejection are because of HLA antigens
- Each individual has their own HLA-antigens
- Identical twins have almost similar HLA antigens

Organ Transplantation

- Like RBC's, other cells and tissues of the body, also have antigens that can cause transfusion reactions when transplanted to other individuals
- Such immune reactions that occur include:
 - A. Autograft:
 - Transplantation from one part to another part of the body in the same person
 - B. Isograft → transplant between identical twins

Organ Transplantation

- C. Allograft** → transplant between the same species (from one person to another)
- D. Xenograft** → transplantation between different species (from an animal of one species to that of another)